A HYBRID VISION TRANSFORMER - CNN FRAMEWORK FOR EXPLAINABLE MULTI-DISEASE RETINAL SCREENING

MAJOR PROJECT REPORT submitted in partial fulfillment of the requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

By

ABHINAYA CHALAMALASETTI - 228W1A1275

RAMYA DOVA - 228W1A1284

Under the Guidance of

DR. CH. SUBBA REDDY

ASSISTANT PROFESSOR

DEPARTMENT OF INFORMATION TECHNOLOGY

V R SIDDHARTHA ENGINEERING COLLEGE

(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)

Approved by AICTE &Accredited by NBA

KANURU, VIJAYAWADA-520007

ACADEMIC YEAR

(2025-26)

V.R.SIDDHARTHA ENGINEERING COLLEGE

(Affiliated to JNTUK: Kakinada, Approved by AICTE, Autonomous)
(An ISO certified and NBA accredited institution)
Kanuru, Vijayawada – 520007

CERTIFICATE

This is to certify that this project report titled **"A HYBRID VISION TRANSFORMER - CNN FRAMEWORK FOR EXPLAINABLE MULTI-DISEASE RETINAL SCREENING"** is a bonafide record of work done by **ABHINAYA CHALAMALASETTI (228W1A1275) and RAMYA DOVA (228W1A1284)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

(Dr. Ch. Subba Reddy)
Assistant Professor
Dept. of Information Technology

(Dr. M. Suneetha)
Dean R&D, Professor &Head
Dept. of InformationTechnology

EXTERNAL EXAMINER SIGNATURE

Date of examination:

DEPARTMENT OF INFORMATION TECHNOLOGY

V.R.SIDDHARTHA ENGINEERING COLLEGE

PROJECT SUMMARY

S.No	Item	Description
1	Project Title	A Hybrid Vision Transformer - Cnn Framework for Explainable Multi-Disease Retinal Screening
2	Student Names & Numbers	Chalamalasetti Abhinaya(228W1A1275) Dova Ramya (228W1A1284)
3	Name of The Guide	Dr. Ch. Subba Reddy , Assistant Professor
4	Research Group	Deep Learning for Medical Imaging
5	Application Area	Health Care
6	Aim of the Project	To develop a hybrid deep learning-based system that integrates Convolutional Neural Networks and Vision Transformers for accurate, automated, and explainable multi-disease retinal classification using fundus images.
7	Project Outcomes	• Development of an automated system for detecting multiple retinal diseases from fundus images • Improved diagnostic accuracy through hybrid CNN–Transformer architecture • Ability to classify multiple diseases simultaneously (multi-label classification) • Enhanced feature representation using cross-attention and feature recalibration

Student Signatures

1.
2.

Signature of the Guide

ACKNOWLEDGEMENT

First and foremost, I sincerely salute our esteemed institution **V.R SIDDHARTHA ENGINEERING COLLEGE** for giving me this opportunity for fulfilling my project. I am grateful to our Principal **Dr. A.V.RATNA PRASAD**, for his encouragement and support all through the way of my project.

On the submission of this Project report, I would like to extend my honour to **Dr. M.Suneetha**, Head of the Department, IT for her constant motivation and support during the course of my work.

I feel glad to express my deep sense of gratefulness to my project guide **Dr. Ch. Subba Reddy, Assistant Professor** for **his** guidance and assistance in completing this project successfully.

I would also like to convey my sincere indebtedness to all faculty members, including supporting staff of the Department, friends and family members who bestowed their great effort and guidance at appropriate times without which it would have been very difficulty on my part to finish the project work.

TABLE OF CONTENTS

ACKNOWLEDGEMENT	iv
LIST OF FIGURES	vi
LIST OF TABLES	vii
LIST OF SYMBOLS	viii
ABSTRACT	ix
CHAPTER-1 Introduction	**(1-6)**
1.1 Origin of the Problem	01
1.2 Basic definitions and Background	01
1.3 Problem Statement with Objectives and Outcomes	03
1.4 Related Applications	05
CHAPTER-2 Review of Literature	**(7-15)**
2.1 Description of Existing Systems	07
2.2 Summary of Literature Study	11
2.3 Software Requirements Specification	14
CHAPTER-3 Proposed Method	**(16-34)**
3.1 Design Methodology	16
3.2 System Architecture Diagram	17
3.3 Description of Algorithms	21
3.4 Description of datasets, Requirements and Tools	32
CHAPTER-4 Results and Observations	**(35-61)**
4.1 Overview of Experiments	35
4.2 Results of Study 1	35
4.3 Results of Study 2	43
4.4 Results of Proposed Model	51
CHAPTER-5 Conclusion and Future work	**(62-65)**
5.1 Conclusion	62
5.2 Future Study	64
REFERENCES	x
APPENDIX	xii

LIST OF FIGURES

Figure	Pg No
Figure 3.2.1 Architecture Diagram	17
Figure 4.2.5.1 Confusion Matrix	41
Figure 4.3.3.1 Classification Report of Ensemble Model	45
Figure 4.3.4.1 Confusion matrix of Ensemble model	47
Figure 4.3.5.1.Micro Averaged ROC curve	49
Figure 4.3.5.2 ROC-AUC score comparison	49
Figure 4.4.1.1 Training and Vadilation curves over epochs	52
Figure 4.4.3.1 ROC curves per disease category on the test set	54
Figure 4.4.4.1 Normalized confusion matrices for each disease category	55
Figure 4.4.6.1 Grouped Bar chart of per-disease AUC,F1 etc	57
Figure 4.4.7.1 Grad-CAM saliency maps	58

LIST OF TABLES

Table	Pg No
Table 2.2.1 Literature Summary	11
Table 3.4.4.1 Distribution of images over classes	33
Table 4.2.1.1 Training and validation accuracy over epochs	36
Table 4.2.2.1 Training and validation loss over epochs	37
Table 4.2.4.1 Classification Report on Test-set	40
Table 4.3.1.1 Cross-Validation Fold Results	43
Table 4.3.2.1 Test Accuracy and AUC Scores	44
Table 4.3.5.2.2 AUC Comparison Table	50
Table 4.4.2.1 Per-disease classification metrics on the test set	52
Table 4.4.5.1 Macro-averaged and weighted-averaged precision, recall, and F1 scores	56
Table 4.5.1 Model Performance Comparison	60

LIST OF SYMBOLS

Equation No	Brief Description	Pg No
Eq-(1)	Accuracy	39
Eq-(2)	Precision	39
Eq-(3)	Recall	39
Eq-(4)	F1-Score	39
Eq-(5)	Specificity	39

ABSTRACT

Retinal diseases are a major cause of vision impairment worldwide, and early detection is essential for effective treatment and prevention of blindness; this project presents a hybrid deep learning framework for automated multi-disease retinal classification using fundus images by combining the strengths of Convolutional Neural Networks and Vision Transformers to capture both local and global features, where EfficientNet-B4 is utilized for extracting fine-grained spatial features and the Vision Transformer models global contextual relationships, with a bidirectional cross-attention mechanism employed to fuse these features followed by squeeze-and-excitation based feature recalibration to enhance representation; the model is trained on a multi-label retinal dataset containing six disease categories including diabetic retinopathy, age-related macular degeneration, macular hole, diabetic neuropathy, myopia, and tessellated fundus, and preprocessing techniques such as contrast enhancement and illumination correction are applied to improve image quality while advanced training strategies including data augmentation, class imbalance handling, and test-time augmentation are used to improve generalization; experimental results demonstrate that the proposed model achieves strong performance with a mean AUC of approximately 0.91 across all classes, showing high accuracy for several diseases particularly macular hole and myopia, while performance for underrepresented classes highlights the challenges associated with class imbalance; furthermore, explainability is incorporated using Grad-CAM, which provides visual insights into model predictions and confirms that the model focuses on clinically relevant regions, thereby improving trust and interpretability; overall, the proposed hybrid framework effectively enhances multi-disease retinal classification by combining complementary feature representations and providing interpretable results, making it suitable for automated screening and clinical decision support applications.

Keywords: Retinal Disease Classification, Multi-label Learning, EfficientNet-B4, Vision Transformer, Hybrid Deep Learning, Cross-Attention, Explainable Artificial Intelligence, Grad-CAM, Fundus Imaging

CHAPTER-1

INTRODUCTION

In this chapter, an introduction to the proposed study on multiple diseases' retinal classifications through a hybrid approach of deep learning is presented. The first step is the identification of the problem along with the necessity for automated detection of retinal diseases. In addition, some basic concepts related to the problem domain will be explained like retinal images, CNNs, and Vision Transformers. Furthermore, problem definition, objectives, and expected outputs will be elaborated upon. Lastly, various practical applications of the proposed system will be discussed.

1.1 Origin of the Problem

Some common retinal disorders include diabetic retinopathy (DR), age-related macular degeneration (ARMD), macular hole (MH), diabetic neuropathy (DN), myopia (MYA), and tessellated fundus (TSLN). These disorders contribute substantially to visual impairment and blindness cases worldwide. Accurate diagnosis and early detection are key to preventing any permanent damage to the patient's eyesight. Nonetheless, manual analysis of retinal fundus images demands considerable time and skilled personnel that may not be available in remote areas [1].

The global prevalence of diabetes and aging individuals has raised an urgent need for effective diagnostics systems. The conventional method highly depends on human analysts and thus cannot be considered objective or accurate. In this regard, researchers have considered the use of artificial intelligence and deep learning approaches to address this challenge.

In recent years, computer vision has made tremendous strides in building automated retinal disorder classification tools [2]. However, current models exhibit significant challenges in detecting several diseases at once because of differences in retinal fundus images. Therefore, there arises a pressing need for an efficient system that can handle multi-disease classification tasks.

1.2 Basic Definitions and Background

1.2.1 Retinal Fundus Image:

Fundus retinal photographs are colorful images captured using a fundus camera of the inside part of the eye. These offer a good view of the physiological components, including the retina, optic disc, macula, fovea, and retinal blood vessels, which are important elements of vision and whose defects may point out any problems that one is experiencing. Fundus photography has gained wide application because of its non-invasive nature, and changes in

color, texture, and structure in the fundus retinal photographs have been important in detecting diseases like diabetic retinopathy, macular degeneration, and myopic complications [3].

1.2.2 Convolutional Neural Networks (CNNs):

Convolutional Neural Networks (CNNs) are a form of deep learning neural networks that can analyze grid structured data (such as image data). The architecture of a CNN consists of a fully connected layer, a pooling layer, and a convolutional layer. The application of filters with the convolutional layer serves to extract useful features such as edge, texture, and shape of the input image data [4].

As the dimension of the output volume is reduced by pooling layers, this also reduces the amount of computation required for processing image data with CNNs. Because they can extract a series of hierarchical based representations from simple edges to complicated disease-specific patterns, CNNs are quite effective for analyzing medical images. In addition, they are widely used for analyzing retinal images and detecting exudates, hemorrhage, and microaneurysms.

1.2.3 Vision Transformers (ViTs):

Vision Transformers (ViTs) are relatively new neural network architectures based on transformers, initially introduced for language processing [5]. The main difference between vision transformers and convolutional networks is that while the latter processes images with localized kernels, in ViTs, an image is split into patches, and each patch is treated like a token. Tokens in vision transformers are processed by attention mechanisms, allowing the network to establish connections between the patches and thus learn the context of the image globally. For instance, for the task of retinal image classification, knowing the connection between various parts of the retina could be essential. However, vision transformers usually need large amounts of data for proper training and do not detect local details as efficiently as convolutional networks.

1.2.4 Hybrid CNN-Transformer Model:

A **hybrid CNN-Vision Transformer** architecture takes advantage of the strengths of both types of neural networks. In such architectures, the CNN part extracts local features, while the Vision Transformer part extracts global features from the inputs. The combined approach enables the system to take into account both detailed and structural features, hence

improving the efficiency of solving tasks that require deep learning abilities, such as multiple disease detection on a retinal image [6].

1.2.5 Attention Mechanisms:

Attention mechanism is another essential part of advanced deep learning algorithms. Attention helps the algorithm to pay attention to important elements in the input and ignore less significant ones. The use of the attention mechanism in the analysis of images helps to identify areas of the image where there might be a presence of disease-related characteristics [7]. The cross-attention mechanism allows the interaction between various representations of features.

1.2.6 Explainable Artificial Intelligence (XAI):

Explainable Artificial Intelligence (XAI) is an approach used in deep learning to ensure the transparency of the process that leads to decisions made. It plays an important role in medicine since doctors should be convinced of the results that have been obtained through the use of automation tools. The Grad-CAM method produces heatmaps showing the parts of an image that have affected the result the most. Such a visualization can be useful to make sure that the model focuses on clinically significant areas, like lesions in retinal images [8].

1.2.7 Image Preprocessing:

Pre-processing of images is important in making deep learning models more efficient. Fundus retina images often have problems such as insufficient lighting, low contrasts, and presence of noises. Methods like contrast enhancement and illumination normalization are among the popular methods used during pre-processing [9]. Adaptive histogram equalization makes it easier for detailed information to be identified from an image. This method can help improve visualization. On the other hand, illumination normalization aims to normalize images.

1.3 Problem Statement with Objectives and Outcomes

1.3.1 Problem Statement:

Eye diseases contribute to the majority of visual disorders and blindness around the world. For example, diabetic retinopathy (DR), age-related macular degeneration (ARMD), macular hole (MH), diabetic neuropathy (DN), myopia (MYA), and tessellated fundus (TSLN) usually occur slowly without showing any signs during the initial stages. Consequently, it is difficult to detect these diseases promptly, resulting in late diagnosis and vision loss.

However, the advancements in AI and deep learning have enabled the creation of automated approaches that can detect retinal diseases. There are several problems that exist with the existing automated approaches used. For instance, some approaches only work with individual diseases and therefore, fail when several diseases co-exist within a single retinal image. The resemblance of the retinal diseases makes it difficult for such an approach to differentiate.

One of the other challenges associated with automated methods is the poor quality of the images. Retinal images may be subjected to various problems like uneven illumination, poor contrast, and noisy images. Deep learning approaches like convolutional neural networks mainly focus on the local feature extraction without incorporating global information in the image while vision transformers do so.

Moreover, most current models function as "black boxes," where decisions are made without any explanation behind them [9]. This problem becomes even more critical when applying these models to the domain of medicine since it lowers clinicians' confidence in such tools. In light of this, there is a requirement for an effective and interpretable solution capable of recognizing several retinal disorders using fundus images.

1.3.2 Objectives:

The main goal of this research is to design a novel deep learning model that can perform multi-disease classification on retinal images. The detailed goals are given below:

1. To design a hybrid deep learning architecture based on Convolutional Neural Networks and Vision Transformers.
2. To design an architecture that can learn both the local and global features of the retinal images.
3. To use attention mechanism-based feature fusion in the architecture.
4. To design a multi-label classification model for retinal images where a single image can have more than one disease.
5. To perform contrast enhancement, illumination correction and other preprocessing on retinal images to enhance the quality.
6. To tackle the problem of imbalanced data in classification models.
7. To use data augmentation, MixUp and CutMix for training the deep learning model to improve its performance.
8. To perform model interpretation using explainable AI techniques.

9. To compare the performance of the designed model with that of baseline models like CNN-based model, ViT-based model and hybrid model without attention mechanism.
10. To compare the proposed model with baseline models such as CNN-only, ViT-only, and hybrid models without attention mechanisms.

1.3.3 Outcomes:

The expected outcomes of this project are outlined below:

1. Development of a reliable and efficient automated system for multi-disease retinal classification.
2. Improved classification performance compared to traditional single-architecture models due to the integration of CNN and Transformer features.
3. Enhanced feature representation through the use of attention mechanisms and hybrid architecture.
4. Better generalization capability achieved through preprocessing and data augmentation techniques.
5. Accurate identification of multiple retinal diseases within a single image.
6. Generation of interpretable results through visual explanations, increasing trust in the system.
7. Reduction in diagnostic workload for ophthalmologists by providing automated screening support.
8. Potential applicability in real-world healthcare settings, including telemedicine and large-scale screening programs.

1.4 Related Applications

The proposed system has several practical applications in the field of healthcare and medical imaging:

1. **Automated Screening Systems:** Can be used in hospitals and clinics for early detection of retinal diseases, reducing the workload on ophthalmologists.
2. **Telemedicine Applications:** Enables remote diagnosis and monitoring of patients, especially in rural and underserved areas.
3. **Clinical Decision Support Systems:** Assists medical professionals by providing reliable predictions and visual explanations.
4. **Mass Screening Programs:** Useful for large-scale screening of diabetic patients to detect complications such as diabetic retinopathy.

5. **Research and Medical Studies:** Can be used to analyze retinal datasets and support further research in ophthalmology and artificial intelligence.
6. **Mobile and AI-based Diagnostic Tools:** Integration into portable devices for real-time screening and diagnosis.

CHAPTER-2

REVIEW OF LITERATURE

This chapter describes the review of literature that we have taken from various papers and considered all the points mentioned in the papers.

2.1 Description of Existing Systems

Hemal, M. M., & Saha, S. are proposed "**Explainable deep learning-based meta-classifier approach for multi-label classification of retinal diseases"**. The paper introduces a hybrid deep learning framework for automated detection and classification of multiple retinal diseases. It combines pre-trained models such as DenseNet169 and EfficientNetB3 with a Random Forest meta-classifier to improve classification performance. The model performs multi-label classification on two retinal datasets, achieving high accuracies of 98.1% and 97.7%, respectively, along with strong precision, sensitivity, and specificity. The approach also utilizes preprocessing techniques like image resizing, noise reduction, and data augmentation to enhance image quality and handle data limitations. Additionally, Explainable AI using LIME is incorporated to provide visual interpretation of model predictions, improving transparency and clinical trust. However, the study has some drawbacks, including high computational complexity due to multiple deep learning models, limited validation [10].

Darapaneni, C. M., Babu, B. S., Satyanarayana, S. V., Madupu, L. N. K. S., Sujay, V., Janardhan, N., Kishore, G. D. K., Kaveti, K. K., & Sairam, R. C. S. N. P. are proposed **"Hybrid Vision Transformer-CNN Framework for Automated Retinal Disease Detection from Fundus Images"**. The paper introduces a hybrid deep learning framework that combines Vision Transformers (ViTs) with Convolutional Neural Networks (CNNs) for automated detection and multi-class classification of retinal diseases from fundus images. It classifies diseases such as diabetic retinopathy (DR), glaucoma, and age-related macular degeneration (AMD), achieving a classification accuracy of 96.3%, precision of 95.1%, recall of 96.7%, F1-score of 95.9%, and an AUC score of 0.982. The model utilizes multiple datasets including APTOS, Messidor, and EyePACS, and employs preprocessing techniques such as image resizing, normalization, contrast enhancement using CLAHE, data augmentation, optic disc and vessel segmentation, and green channel extraction to enhance image quality and highlight retinal features. The hybrid CNN–ViT architecture incorporates convolutional

layers for feature extraction and transformer-based attention mechanisms for capturing global dependencies, along with statistical validation using ANOVA ($p < 0.01$) to demonstrate improved performance over existing models. However, the study has drawbacks such as high computational complexity due to the hybrid architecture, challenges related to data imbalance and noise, limited focus on real-time deployment in low-resource environments, and the need for future work to improve efficiency, scalability, and practical clinical implementation [11].

Ding, X., Zhang, Y., Wu, H., & Wang, J. are proposed **"Deep Learning-Based Classification of Fundus Images for Retinal Disease Detection".** The paper introduces a deep learning-based approach for automated detection and classification of retinal diseases using fundus images. It utilizes convolutional neural networks (CNNs) to extract features and classify diseases such as diabetic retinopathy (DR), achieving high accuracy and improved diagnostic performance. The model is trained on publicly available retinal datasets and applies preprocessing techniques like image normalization, resizing, and enhancement to improve image quality and model performance. The system demonstrates strong results in terms of accuracy, sensitivity, and specificity, showing its effectiveness in assisting early diagnosis and clinical decision-making. However, the study has certain drawbacks, including dependency on large annotated datasets, limited generalizability due to dataset constraints, and high computational requirements, along with the need for further validation on diverse real-world datasets and improvements for deployment in resource-constrained environments [12].

Singh, D., Agarwal, S., & Mishra, S. are proposed **"Retinal Fundus Multi-Disease Image Classification using Hybrid CNN-Transformer-Ensemble Architectures"**. The paper introduces a hybrid deep learning model combining CNNs, Transformers, and ensemble techniques for multi-label classification of retinal diseases. It classifies fundus images into 20 disease categories and achieves a model score of 0.9166, outperforming baseline models. The model utilizes the MuReD dataset and applies preprocessing techniques such as Weighted Random Sampling, data augmentation, image resizing, normalization, and enhancement methods to improve performance. It incorporates deep CNN models like DenseNet and ResNet along with Transformer-based architectures such as C-Tran and IEViT for effective feature extraction and global context learning. However, the study has drawbacks including limitations due to small and imbalanced datasets, risk of overfitting, high computational

complexity, and the need for further validation and improvement for real-world clinical deployment [13].

Thamilarasi, R., & Raja, S. are proposed **"Retinal Disease Classification using Deep Learning Techniques" .** The paper introduces a deep learning-based approach for automated classification of retinal diseases using fundus images. It utilizes convolutional neural networks (CNNs) for feature extraction and classification, showing effective performance in detecting retinal conditions. The model is trained on publicly available datasets and applies preprocessing techniques such as image resizing, normalization, and data augmentation to enhance image quality and improve model generalization. The system demonstrates strong results in terms of sensitivity and specificity, making it useful for early disease detection. However, the study has drawbacks including dependence on large labeled datasets, limited generalizability due to dataset constraints, risk of overfitting [14].

Djoumessi, K., Mensah, S. O., and Berens, P. (2025) proposed the paper titled **"A Hybrid Fully Convolutional CNN-Transformer Model for Inherently Interpretable Disease Detection from Retinal Fundus Images" .**The authors introduced a hybrid architecture that combines Convolutional Neural Networks (CNNs) for extracting local features and Vision Transformers (ViTs) for capturing global dependencies through dual-resolution self-attention. A key contribution of this work is its **interpretable-by-design framework**, where class-specific evidence maps are generated directly without relying on post-hoc methods such as Grad-CAM. The model was evaluated on benchmark datasets including Kaggle Diabetic Retinopathy and AREDS. The proposed model achieved **high classification performance with accuracy around 85%–87% for diabetic retinopathy detection and up to 91% for AMD classification**, along with strong Cohen's Kappa scores, demonstrating state-of-the-art performance while maintaining interpretability [15].

Elkenawy, E. M., Khodadadi, N., Gaber, K. S., Khodadadi, E., Alhussan, A. A., Khafaga, D. S., and Eid, M. M. (2026) proposed the paper titled **"Automated Retinal Disease Classification Using Deep Learning and AlexNet with Statistical Models Analysis"** . The authors developed an automated deep learning framework for classifying retinal diseases into four categories: Normal, Diabetic Retinopathy, Cataract, and Glaucoma using multiple CNN architectures including EfficientNet-B0, EfficientNet-B7, and AlexNet. Among these, the

AlexNet model achieved the best performance with an accuracy of 93.65%, sensitivity of 94.39%, and specificity of 98.05%, demonstrating its effectiveness in multi-class retinal disease classification. The system also incorporates **statistical validation techniques and SHAP-based explainability**, which highlights important retinal regions such as the optic disc and macula, improving model transparency and clinical trust [16].

Sweidan, S., and Taha, A. (2025) proposed the paper titled **"DeepRetina Framework for Multi-retinal DiseasesClassification"**. The authors introduced a novel **DeepRetina framework** for multiclass retinal disease classification by addressing dataset inconsistency through **data harmonization techniques**. The study unified five different retinal datasets into a large dataset of **29,966 high-resolution fundus images across eight disease categories**, improving model generalization. The framework utilizes deep learning models such as **Custom CNN, EfficientNetV2, and MobileNetV3Large**, where **EfficientNetV2 achieved the best performance with 79% accuracy and 54% precision**. The model also integrates advanced techniques like **Neural Architecture Search (NAS)** and **Adaptive Variance Thresholding (AVT)** to enhance feature selection and classification performance [17].

Wang, D., Lian, J., and Jiao, W. (2024) proposed the paper titled **"Multi-label classification of retinal disease via an ovel vision transformer model"**. The authors introduced a **novel Vision Transformer (ViT)-based model** specifically designed for **multi-label retinal disease classification**, addressing the real-world challenge where a single retinal image may contain multiple diseases simultaneously. Unlike traditional CNN-based methods, the proposed model uses a **self-attention mechanism** to capture global relationships between pixels. The model processes retinal images by dividing them into patches, applying **linear embeddings, positional encoding, multi-head self-attention, and MLP layers**, followed by a multi-label classifier with sigmoid activation. The model was evaluated on the **ODIR-2019 dataset**, which contains eight retinal disease categories, and achieved strong performance with **AUC of 0.938, F1-score of 0.919, and Kappa of 0.645** [18].

Bitto, A. K., Karim, R., Begum, M. H., Khan, M. F. I., Hassan, M. M., and Masum, A. K. M. (2025) proposed the paper **titled "Explainable AI Based Deep Ensemble Convolutional Learning for Multi-Categorical Ocular Disease Prediction"** presents a novel system for automated classification of multiple retinal diseases using deep learning and

explainable AI techniques. The authors developed a deep ensemble model that combines multiple pre-trained convolutional neural networks such as VGG16, DenseNet121, MobileNetV2, and InceptionV3 using transfer learning. The system classifies retinal images into four categories: diabetic retinopathy, glaucoma, cataract, and normal, by integrating predictions through majority voting and probability averaging methods. The proposed approach achieved a high classification accuracy of **90%**, outperforming individual models such as DenseNet121 (87.91%) and MobileNetV2 (86.01%), thereby demonstrating the effectiveness of ensemble learning. Additionally, the model incorporates Explainable AI techniques like Grad-CAM to highlight important regions in retinal images, improving interpretability and clinical trust. However, the study has some limitations. It relies on a single dataset, which may affect the generalizability of the model in real-world scenarios. The use of multiple deep learning models increases computational complexity, making it less suitable for low-resource environments. Furthermore, misclassification between visually similar diseases such as cataract and glaucoma indicates the need for improved feature extraction and model optimisation [19].

2.2 Summary of Literature Study

The following Table 2.2.1 gives the summary of our literature review done.

Table 2.2.1 : Literature Summary

Title and Author Name(s)	**Year**	**Work Done**	**Algorithms**	**Dataset Used**	**Conclusion**	**Limitations**
"Automated Retinal Disease Classification Using Deep Learning and AlexNet with Statistical Models Analysis" (E. M. Elkenawy, N. Khodadadi, K. S. Gaber, E. Khodadadi, A. A. Alhussan, D. S. Khafaga, M. M. Eid)[10]	2026	Multi-class retinal disease classification using deep learning	AlexNet, EfficientNet	Retinal datasets	Achieved **93.65% accuracy**, high sensitivity and specificity	Older architecture limitations, dataset dependency

"Explainable AI Based Deep Ensemble Convolutional Learning for Multi-Categorical Ocular Disease Prediction" (A. K. Bitto, R. Karim, M. H. Begum, M. F. I. Khan, M. M. Hassan, A. K. Masum)[11]	2025	Deep ensemble model with explainable AI	CNN Ensemble (VGG16, DenseNet121, MobileNetV2, InceptionV3)	NKS Eye Disease dataset	Achieved **90% accuracy**, improved interpretability	High computationalcost, single dataset limitation
"DeepRetina Framework for Multi-retinal Diseases Classification" (S. Sweidan, A. Taha)[12]	2025	Multi-disease classification using data harmonization	CNN, EfficientNetV2, MobileNetV3	Combined datasets (29,966 images)	Achieved **79% accuracy**, improved generalization	Class imbalance, lower accuracy
"A Hybrid Fully Convolutional CNN-Transformer Model for Inherently Interpretable Disease Detection from Retinal Fundus Images" (K. Djoumessi, S. O. Mensah, P. Berens)[13]	2025	Hybrid CNN + Transformer model with interpretability	CNN + Vision Transformer	Kaggle DR, AREDS	Achieved **85–87% (DR), up to 91% (AMD)**	High computational cost, complex architecture
"Hybrid Vision Transformer-CNN Framework for Automated Retinal Disease Detection from Fundus Images" (C. M. Darapaneni, B. S. Babu, S. V. Satyanarayana, L. N. K. S. Madupu, V. Sujay, N. Janardhan, G. D. K. Kishore, K. K. Kaveti, R. C. S. N. P. Sairam) [14]	2025	Developed hybrid CNN + ViT model for multi-class retinal disease detection	CNN + Vision Transformer	APTOS, Messid, EyePAC	Achieved 96.3% accuracy, strong F1-score and AUC	High complexity, data imbalance, not suitable for low-resource

"Explainable deep learning-based meta-classifier approach for multi-label classification of retinal diseases" (M. M. Hemal, S. Saha)[15]	2025	Proposed a multi-label retinal disease classification framework using a meta-classifier integrated with explainable AI techniques.	Deep CNN models (DenseNet169, EfficientNet B3) with Random Forest meta-classifier	Multi-label retinal fundus datasets	Achieved 98.1% and 97.7% accuracy, improved precision, sensitivity, and specificity	High computational complexity, limited real-world validation
"Deep Learning-Based Classification of Fundus Images for Retinal Disease Detection" (X. Ding, Y. Zhang, H. Wu, J. Wang)[16]	2024	Automated retinal disease classification using deep learning	CNN	Public retinal datasets	High accuracy with improved diagnosis performanc	Needs large dataset, limited generalization
"Retinal Fundus Multi-Disease Image Classification using Hybrid CNN-Transformer-Ensemble Architectures" (D. Singh, S. Agarwal, S. Mishra)[17]	2024	Multi-label classification of 20 retinal diseases using hybrid ensemble model	CNN + Transformer + Ensemble	MuReD dataset	Achieved 0.9166 score, better than baseline models	Small dataset, overfitting risk, high computation
"Multi-label classification of retinal disease via a novel vision transformer model" (D. Wang, J. Lian, W. Jiao)[18]	2024	Multi-label classification using Vision Transformer	Vision Transformer	ODIR-2019	Achieved AUC 0.938, F1-score 0.919	Class imbalance, high computation

"Retinal Disease Classification using Deep Learning Techniques" (R. Thamilarasi, S. Raja)[19]	2023	Deep learning-based retinal disease classification	CNN	Public datasets	Good sensitivity and specificity	Needs large 14abelled data, overfitting ris

2.3 Software Requirements Specification

2.3.1 Purpose

This software aims to automate the detection of Diabetic Retinopathy (DR) from retinal fundus images using the EfficientNet-B3 deep learning model. It provides an accurate, efficient, and scalable diagnostic support system to assist ophthalmologists and healthcare providers in early detection and classification of DR into five stages.

2.3.2 Scope

The software processes retinal fundus images, classifies them into one of the five DR categories (No DR, Mild, Moderate, Severe, Proliferative), and visualizes classification results along with performance metrics such as accuracy, confusion matrix, and ROC AUC. It is designed for deployment in eye hospitals, teleophthalmology centers, clinics, and public health screening programs.

2.3.3 Intended Users

- Ophthalmologists
- General physicians
- Medical researchers
- Public health officials
- Healthcare software developers

2.3.4 Functional Requirements

- **FR1:** Upload retinal fundus images (individually or in batches)
- **FR2:** Preprocess and normalize input images using transformations (resize, flip, rotation, normalization)
- **FR3:** Classify images using a pre-trained EfficientNet-B3 model

- **FR4:** Display classification result (No DR, Mild, Moderate, Severe, Proliferative)
- **FR5:** Visualize performance metrics such as accuracy, precision, recall, F1-score, ROC AUC
- **FR6:** Generate and display confusion matrix
- **FR7:** Save classification results and model checkpoints
- **FR8:** Plot training/validation accuracy and loss curves for model performance tracking

2.3.5 Non-Functional Requirements

- **NFR1:** System should return classification results within 2–5 seconds per image
- **NFR2:** Capable of handling high-resolution retinal images efficiently
- **NFR3:** User-friendly interface for image upload and result viewing
- **NFR4:** Ensure secure handling and storage of medical image data
- **NFR5:** Platform-independent deployment (support for Windows/Linux)

2.3.6 Software Requirements

- **Operating System:** Windows or Linux
- **Programming Language:** Python
- **Frameworks and Libraries:**
- PyTorch (deep learning and EfficientNet-B3 model)
- torchvision, timm (model and data utilities)

CHAPTER – 3

PROPOSED METHOD

This chapter describes the overall methodology of the proposed hybrid deep learning framework for multi-disease retinal classification. It explains the design approach, system architecture, algorithms used, and details about the dataset, tools, and requirements. The proposed system integrates preprocessing techniques, a hybrid CNN–Vision Transformer model, and explainability methods to achieve accurate and reliable predictions.

3.1 Design Methodology

The proposed system follows a structured pipeline to perform multi-disease retinal classification. The methodology is designed to ensure accurate feature extraction, robust learning, and reliable prediction.

The overall workflow of the system consists of the following stages:

3.1.1 Image Acquisition:

Retinal fundus images are collected from a publicly available dataset. These images contain various retinal diseases and corresponding labels.

3.1.2 Preprocessing:

The input images undergo preprocessing to enhance their quality. Techniques such as contrast enhancement and illumination correction are applied to highlight important features such as lesions and blood vessels. This step reduces noise and improves model performance.

3.1.3 Data Augmentation:

To increase dataset diversity and prevent overfitting, augmentation techniques such as flipping, rotation, scaling, and brightness adjustments are applied. This helps the model generalize better to unseen data.

3.1.4 Feature Extraction:

The system uses a hybrid approach combining:

- CNN (EfficientNet) for extracting local features
- Vision Transformer (ViT) for capturing global context

3.1.5 Feature Fusion using Attention:

The extracted features from both branches are combined using a bidirectional cross-attention mechanism. This allows interaction between local and global features, improving representation.

3.1.6 Feature Refinement:

A squeeze-and-excitation (SE) block is used to recalibrate feature importance by assigning weights to different channels.

3.1.7 Classification:

The refined features are passed through a fully connected classifier to predict multiple diseases.

3.1.8 Explainability:

Grad-CAM is used to generate heatmaps that highlight important regions in the retinal image.

3.1.9 Evaluation:

The model is evaluated using metrics such as accuracy, precision, recall, F1-score, and AUC.

This systematic design ensures that the model effectively captures both fine details and overall structure in retinal images.

3.2 System Architecture Diagram

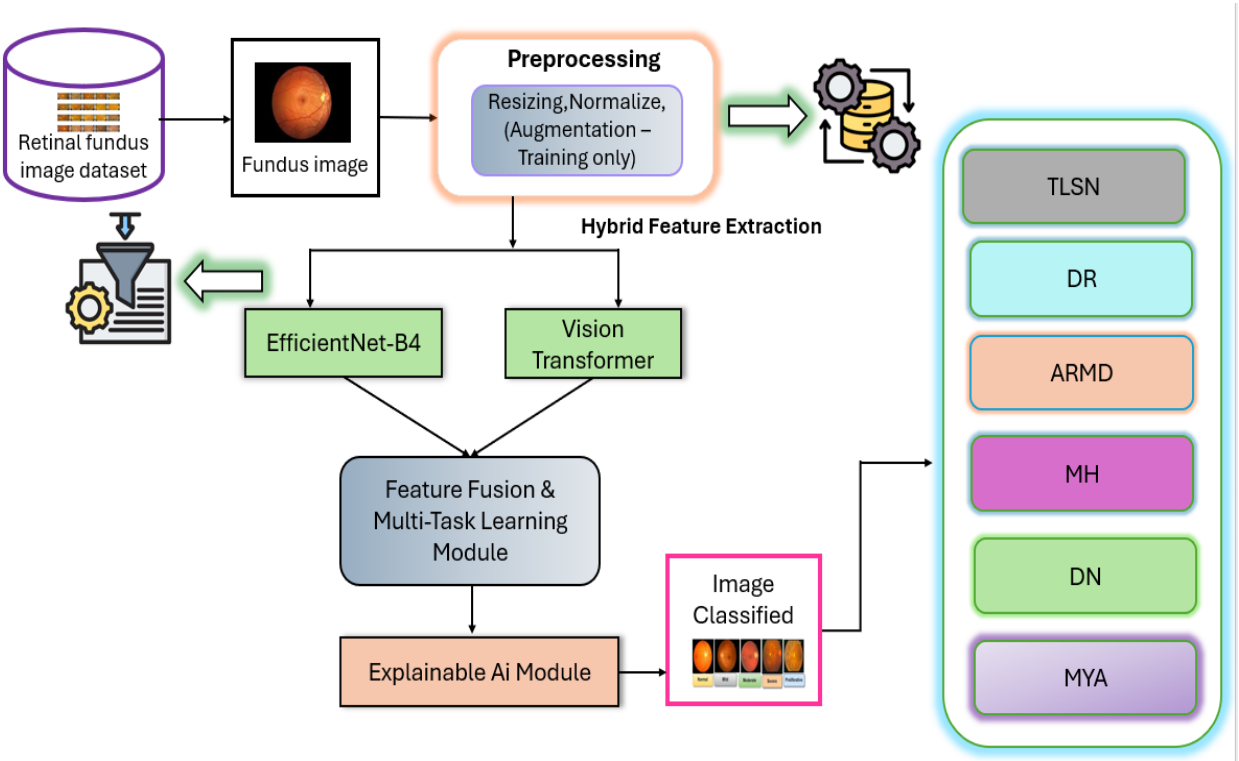

Figure 3.2.1: Architecture Diagram

The proposed system architecture, shown in Figure 3.2.1 is designed to perform multi-disease retinal classification using a hybrid deep learning framework. It consists of multiple interconnected

modules that process retinal fundus images step-by-step, transforming raw input into meaningful predictions along with visual explanations.

The architecture is composed of the following major components:

1. **Input Layer :**

   The system begins with the input of retinal fundus images. These images are collected from a structured dataset and may vary in resolution, lighting conditions, and quality.

   - Images are typically in RGB format
   - Contain important retinal structures such as optic disc, macula, and blood vessels
   - May include noise, uneven illumination, and artifacts

   The input layer serves as the entry point for the entire processing pipeline.

2. **Preprocessing Module :**

   The preprocessing module enhances the quality of input images to make disease-related features more prominent.

   This module performs:

   - **Contrast Enhancement:** Improves visibility of lesions and fine structures
   - **Illumination Correction:** Reduces uneven brightness across the image
   - **Noise Reduction:** Removes unwanted distortions
   - **Region Focus:** Emphasizes the retinal region and suppresses background

   The output of this stage is a clean and enhanced image that is more suitable for feature extraction.

3. **Data Augmentation Module**

   To improve model generalization, augmented versions of the input images are generated during training.

   This includes:

   - Flipping (horizontal and vertical)
   - Rotation and scaling
   - Brightness and contrast variation
   - Noise addition and blurring

   These transformations simulate real-world variations and help the model learn robust features.

4. **CNN Feature Extraction (EfficientNet-B4)**

   The preprocessed image is passed through the CNN branch based on EfficientNet-B4.

   Key functions:

   - Extracts **local features** such as edges, textures, and lesions

- Learns hierarchical representations from low-level to high-level features
- Efficient architecture reduces computational cost

This branch is particularly effective in identifying fine-grained abnormalities in retinal images.

5. Transformer Feature Extraction (Vision Transformer)

In parallel, the same input image is passed through the Vision Transformer branch.

Key functions:

- Divides the image into patches
- Applies self-attention to capture relationships between patches
- Learns **global contextual information**

This branch helps the model understand the overall structure and spatial relationships in the image.

6. Feature Projection Layer

Since the CNN and Transformer outputs have different dimensions, a projection layer is used.

Key Functions:

- Converts features into a common feature space
- Aligns representations for effective fusion
- Applies normalisation for stable learning

This ensures compatibility between both feature sets.

7. Cross-Attention Fusion Module

This is the key component of the proposed system.

Functions:

- Allows CNN features to attend to Transformer features
- Allows Transformer features to attend to CNN features
- Enhances interaction between local and global representations

This bidirectional attention mechanism helps the model focus on the most relevant features by combining complementary information.

8. Feature Fusion Layer

The refined features from both branches are combined.

- Concatenation of feature vectors
- Formation of a unified representation

This fused feature contains both detailed and contextual information.

9. Feature Recalibration (SE Block)

A squeeze-and-excitation (SE) block is applied to refine the fused features.

Functions:

- Assigns importance weights to feature channels
- Enhances relevant features
- Suppresses less important information

This improves the quality of feature representation before classification.

10. Classification Layer

The refined features are passed through a fully connected neural network.

Functions:

- Maps features to disease classes
- Produces logits for each class
- Applies sigmoid activation for multi-label prediction

Output:

- Probability scores for each disease
- Final classification based on thresholds

11. Loss and Optimization Module

During training, the model is optimized using a combination of loss functions.

Functions:

- Compute prediction error
- Handle class imbalance
- Improve learning on difficult samples

Optimization techniques such as AdamW and learning rate scheduling are used to update model weights.

12. Evaluation Module

The model's performance is evaluated using multiple metrics:

- Accuracy
- Precision
- Recall
- F1-score
- AUC-ROC

This ensures a comprehensive assessment of classification performance.

13. Explainability Module (Grad-CAM)

The final module provides visual explanations of model predictions.

Functions:

- Generates heatmaps highlighting important regions
- Shows where the model is focusing
- Helps validate predictions

This improves interpretability and trust in the system.

3.3 Description of the Algorithms

The proposed system integrates multiple algorithms and techniques to perform accurate and robust multi-disease retinal classification. Each component plays a specific role in improving performance, handling data challenges, and ensuring reliable predictions. The major algorithms used in the system are described below.

3.3.1 Image Preprocessing Algorithm

Image preprocessing is performed to enhance the quality of retinal fundus images and improve feature visibility. Since retinal images often suffer from uneven illumination and low contrast, preprocessing ensures that important structures such as blood vessels and lesions are clearly highlighted.

The preprocessing pipeline consists of the following steps:

1. Convert the input image from RGB color space to LAB color space.
2. Apply adaptive histogram equalization to enhance contrast.
3. Perform illumination correction using Gaussian-based filtering.
4. Apply a circular mask to focus on the retinal region and remove background noise.
5. Normalize pixel values to maintain consistency across images.

The output of this stage is an enhanced image with improved clarity and reduced noise, which is then passed to the next stage.

3.3.2 Data Augmentation Algorithm

Data augmentation is used to artificially increase the diversity of the training dataset and improve the model's generalization capability. It helps the model learn invariant features under different transformations.

The augmentation process includes:

1. Random horizontal and vertical flipping of images.
2. Rotation of images at various angles.
3. Scaling and shifting of images.
4. Adjustment of brightness, contrast, and color properties.
5. Addition of noise and blur effects.
6. Random occlusion using dropout techniques.

These transformations are applied randomly during training, ensuring that the model is exposed to a wide range of variations. This reduces overfitting and improves performance on unseen data.

3.3.3 Feature Extraction Algorithm

3.3.3.1 EfficientNet-B3

EfficientNet-B3 is a convolutional neural network architecture designed using a compound scaling method that uniformly scales network depth, width, and resolution. Unlike traditional CNNs that arbitrarily increase network size, EfficientNet uses a balanced approach to improve performance while maintaining computational efficiency. The B3 variant is a mid-sized model that provides a good trade-off between accuracy and computational cost.

In medical image analysis, EfficientNet-B3 is particularly effective due to its ability to extract hierarchical features from images. It captures both low-level features such as edges and textures, as well as high-level semantic features. This makes it suitable for detecting fine-grained patterns in retinal images, such as small lesions or abnormalities. Additionally, the use of pretrained weights allows the model to leverage knowledge learned from large datasets, improving performance even with limited medical data.

3.3.3.2 DenseNet121

DenseNet121 is a convolutional neural network architecture characterized by dense connections between layers. In this architecture, each layer receives input from all preceding layers and passes its output to all subsequent layers. This design promotes feature reuse and improves the flow of information and gradients throughout the network.

One of the key advantages of DenseNet121 is its ability to reduce the vanishing gradient problem, making training more stable. It also requires fewer parameters compared to traditional CNNs because it avoids redundant feature learning. In medical imaging tasks, DenseNet121 is effective in capturing complex patterns and maintaining detailed information, which is essential for accurate disease detection.

3.3.3.3 EfficientNet-B5

EfficientNet-B5 is a larger variant in the EfficientNet family, designed to achieve higher accuracy by increasing input resolution and network depth. It requires more

computational resources compared to B3 and B4 but provides improved feature extraction capability.

In retinal image analysis, EfficientNet-B5 can capture very fine details due to its higher resolution input size. This makes it useful for detecting subtle abnormalities that may not be easily visible. However, the increased computational cost can be a limitation in real-time or resource-constrained environments. Despite this, EfficientNet-B5 is often used in research settings where achieving maximum accuracy is a priority.

3.3.3.4 EfficientNet-B4

EfficientNet-B4 is a more advanced version of the EfficientNet family that offers higher accuracy by increasing network capacity through compound scaling. It is deeper and wider than earlier variants, enabling it to learn more complex feature representations.

In this project, EfficientNet-B4 serves as the CNN backbone for extracting local features from retinal fundus images. It is particularly effective in identifying small and subtle disease-related patterns such as microaneurysms, hemorrhages, and exudates. The model's architecture allows it to preserve important spatial details while progressively learning abstract representations.

3.3.3.5 Vision Transformer(ViT)

The Vision Transformer is a deep learning architecture that applies transformer-based models, originally developed for natural language processing, to image analysis. Instead of using convolutional filters, ViT divides an image into smaller patches and treats each patch as a token. These tokens are processed using self-attention mechanisms, allowing the model to learn relationships between different regions of the image.

The main strength of Vision Transformers lies in their ability to capture global context. In retinal images, this is particularly useful for understanding spatial relationships between different anatomical structures. However, ViTs typically require large amounts of data for effective training and may not capture fine local details as efficiently as CNNs.

3.3.3.6 Hybrid Model (EfficientNet-B4 + Vision Transformer)

The proposed hybrid model combines EfficientNet-B4 and Vision Transformer to leverage the strengths of both architectures. EfficientNet-B4 is responsible for

extracting local features such as edges, textures, and small lesions, while the Vision Transformer captures global relationships and contextual information across the entire image.

To effectively integrate these features, a cross-attention mechanism is used. This allows the model to learn interactions between local and global representations, resulting in a more comprehensive understanding of the retinal image. Additionally, a squeeze-and-excitation (SE) block is applied to recalibrate feature importance by emphasizing relevant features and suppressing less important ones.

This hybrid approach addresses the limitations of individual models by combining their advantages. It improves the model's ability to detect multiple diseases simultaneously and enhances overall classification performance. Furthermore, the integration of explainability techniques ensures that the model's predictions are interpretable and aligned with clinical knowledge.

Algorithm 1: EfficientNet-B3 Training

Input: Fundus images X, labels Y

Output: Trained EfficientNet-B3 model

Steps

1. Resize each image to **300×300** and normalize using ImageNet mean/std.

2. Apply augmentations (flip, rotation, color jitter).

3. Initialize EfficientNet-B3 with pretrained weights.

4. Replace classifier with a 5-class fully connected layer.

5. Define loss as **Cross-Entropy**:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{C} y_i log(\hat{y}_i)$$

6. Use **AdamW** optimizer with learning-rate scheduler.

7. For each epoch:

a. Forward pass → compute logits z.

b. Convert logits to probabilities using softmax:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}}$$

c. Compute loss $\mathcal{L}_{CE}$.

d. Backpropagate and update weights.

e. Evaluate model on validation set.

8. Save checkpoint with highest validation accuracy.

Algorithm 1(Inference): EfficientNet-B3

Steps

1. Preprocess input image (resize 300×300, normalize).

2. Forward pass through EfficientNet-B3→ obtain logits.

3. Apply softmax to get class probabilities:

$$\hat{y} = \text{softmax}(z)$$

4.Predict class = argmax ($\hat{y}$).

Algorithm 2: Densenet121 Training

Input: Fundus images X, labels Y

Output: Trained DenseNet121 model

Steps

1. Resize each image to **224×224** and normalize.
2. Apply augmentations to training images.
3. Initialize DenseNet121 with pretrained ImageNet weights.
4. Replace classifier with a 5-class output layer.
5. Use **Cross-Entropy Loss**:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{C} y_i log(\hat{y}_i)$$

6. Optimize using Adam optimizer with learning-rate scheduler.
7. For each epoch:

   a. Perform forward pass → obtain logits z.

   b. Apply softmax to compute $\hat{y}$.

   c. Compute loss and backpropagate.

   d. Update parameters.

   e. Validate on validation set.
8. Save best model based on validation accuracy.

Algorithm 2 (Inference): DenseNet121

Steps

1. Resize image to 224×224 and normalize.
2. Forward pass through DenseNet121.
3. Apply softmax:

$$\hat{y} = \frac{e^z}{\sum e^z}$$

4. Predict class with highest probability.

Algorithm 3: Confidence-Based Ensemble (Final Prediction)

Input: Probabilities from EfficientNet-B5 (P_E) and DenseNet121 (P_D)

Output: Final predicted class

Steps

1. Compute weighted fusion:

$$P_F = w_1 P_E + w_2 P_D, w_1 + w_2 = 1$$

2. Compute confidence of fused output:

$$c = \max(P_F)$$

3. If $c \geq \tau$, set final probability $P_{final} = P_F$.
4. Else choose the model with higher confidence:

$$P_{final} = \begin{cases} P_E, & \text{if } \max(P_E) \geq \max(P_D) \\ P_D, & \text{otherwise} \end{cases}$$

5. Predict class:

$$y_{pred} = \arg\max(P_{final})$$

Algorithm 3: Ensemble Weight & Threshold Optimization (Validation Phase)

Steps

1. Generate validation probabilities P_E and P_D from both models.
2. Search for optimal (w_1, w_2, τ) using Differential Evolution.
3. For each candidate set, compute fused probability P_F.
4. Compute validation accuracy:

$$Acc = \frac{\text{correct predictions}}{\text{total samples}}$$

5. Select parameters giving maximum accuracy.
6. Use these optimal parameters for final test inference.

Algorithm 4: EfficientNet-B5 Training

Input: Fundus images X, labels Y

Output: Trained EfficientNet-B5 model

Steps

1. Resize each image to **380×380** and normalize using ImageNet mean/std.

2. Apply augmentations (flip, rotation, color jitter).

3. Initialize EfficientNet-B5 with pretrained weights.

4. Replace classifier with a 5-class fully connected layer.

5. Define loss as **Cross-Entropy**:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{C} y_i log(\widehat{y_i})$$

6. Use **AdamW** optimizer with learning-rate scheduler.

7. For each epoch:

a. Forward pass → compute logits z.

b. Convert logits to probabilities using softmax:

$$\widehat{y_i} = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}}$$

c. Compute loss $\mathcal{L}_{CE}$.

d. Backpropagate and update weights.

e. Evaluate model on validation set.

8. Save checkpoint with highest validation accuracy.

Algorithm 1(Inference): EfficientNet-B5

Steps

1. Preprocess input image (resize 380×380, normalize).

2. Forward pass through EfficientNet-B5 → obtain logits.

3. Apply softmax to get class probabilities:

$$\hat{y} = \text{softmax}(z)$$

4.Predict class $= \text{argmax}\ (\hat{y})$.

Algorithm 5: EfficientNet-B4 Training

Input: Fundus images X, labels Y

Output: Trained EfficientNet-B4 model

Steps

1. Resize each image to **224×224** and normalize using ImageNet mean/std.

2. Apply preprocessing (contrast enhancement and illumination correction).

3. Perform data augmentation (flip, rotation, noise).

4. Initialize EfficientNet-B4 with pretrained weights.

5. Replace classifier with a multi label output layer (6 classes).

6. Define loss as using **weighted BCE / Focal Loss.**

7. Use **AdamW** optimizer with cosine learning rate-scheduler.

8 . For each epoch:

a. Forward pass through EfficientNet-B4 → feature extraction

b. Generate logits z

c. Apply Sigmoid Activation:

$$\hat{y}_i = \sigma(z_i)$$

d. Compute Loss.

d. Perform backpropagation.

e. Update model parameters.

f. Evaluate on validation set.

g. Save best- performing model.

Algorithm 6: Vision Transformer (ViT) Training

Input: Fundus images X, labels Y

Output: Trained Vision Transformer model

Steps

1. Resize each image to **224×224** and normalize.
2. Divide image into patches (e.g., 16×16).
3. Flatten patches and convert to embeddings.
4. Add positional encoding.
5. Pass embeddings through Transformer encoder layers.
6. Apply multi-head self-attention.
7. Extract global representation from class token.
8. Pass through classification head.
9. Apply Sigmoid Activation for multi-label output:

$$\hat{y}_i = \sigma(z_i)$$

10. Compute loss and update weights using optimizer

Algorithm 7: Proposed Hybrid Model (EfficientNet-B4 + ViT)

Input: Fundus images X, labels Y

Output: Trained Hybrid CNN-Transformer model

Steps

1. Input retinal image and apply preprocessing (CLAHE + illumination correction).
2. Resize image to $\mathbf{224} \times \mathbf{224}$ and normalize.
3. Apply data augmentation techniques.
4. Pass image through EfficientNet-B4 to extract local features:
$$F_{cnn} = CNN(X)$$
5. Pass image through Vision Transformer to extract global features:
$$F_{vit} = ViT(X)$$
6. Project both features into common dimension
7. Apply bidirectional cross-attention.
8. Concatenate features.
9. Apply SE block:
$$F_{refined} = SE(F_{fusion})$$
10. Pass through fully connected layers.
11. Apply sigmoid activation.
$$\hat{y}_i = \sigma(z_i)$$
12. Compute combined loss (Focal + ASL + BCE).
13. Backpropagate and update weights using AdamW.
14. Apply learning rate scheduling.
15. Phase 1: Train classifier (freeze backbone).
16. Phase 2: Fine-tune entire model.
17. Evaluate on validation set.
18. Save model with best AUC score.

3.4 Description of Datasets, Requirements and Tools

3.4.1 Dataset Description

The dataset used in this project is the Retinal Disease Classification Dataset, which is publicly available on Kaggle, Retinal Disease Classification | Kaggle [20]. This dataset contains retinal fundus images labelled for multiple ocular diseases and is suitable for multi-label classification tasks.

The dataset consists of high-resolution retinal images collected under real-world clinical conditions. Each image is associated with multiple disease labels, indicating the presence or absence of specific retinal conditions. This makes the problem more complex compared to single-label classification, as multiple diseases can coexist in a single image.

3.4.2 Dataset Composition

The dataset is divided into three subsets for training, validation, and testing:

- **Training Set:** 1920 images
- **Validation Set:** 640 images
- **Test Set:** 640 images

This split ensures proper training and unbiased evaluation of the model.

3.4.3 Disease Classes

The dataset includes six retinal disease categories:

- **Diabetic Retinopathy (DR):** A diabetes-related condition affecting retinal blood vessels.
- **Age-related Macular Degeneration (ARMD):** A disease affecting the macula and central vision.
- **Macular Hole (MH):** A small break in the macula leading to vision distortion.
- **Diabetic Neuropathy (DN):** Damage caused due to prolonged diabetes.
- **Myopia (MYA):** A refractive error causing blurred distant vision.
- **Tessellated Fundus (TSLN):** A condition associated with high myopia and retinal thinning.

3.4.4 Class Distribution (Training Set)

The dataset exhibits class imbalance, as the number of samples varies across diseases, shown in Table 3.4.4.1.

Table 3.4.4.1: Distribution of images over classes

Disease	Number of Samples
DR	1376
ARMD	1100
MH	1317
DN	1138
MYA	1101
TSLN	1186

This imbalance can affect model performance, especially for underrepresented classes. To address this, techniques such as weighted loss functions and sampling strategies are used.

3.4.5 Dataset Characteristics

- **Multi-label nature:** A single image may contain multiple diseases.
- **Variability in image quality:** Differences in lighting, contrast, and noise.
- **Real-world data:** Reflects clinical conditions, making the problem more challenging.

These characteristics make the dataset suitable for evaluating robust deep learning models.

3.4.6 Hardware Requirements

The proposed system requires the following hardware configuration:

- **Processor:** Intel Core i5 or higher
- **RAM:** Minimum 8 GB (16 GB recommended)
- **GPU:** NVIDIA GPU (e.g., Tesla T4 or RTX series) for faster training
- **Storage:** At least 10 GB free space for dataset and model files

The use of GPU significantly reduces training time and improves computational efficiency.

3.4.7 Software Requirements

The implementation of the proposed system is carried out using the following software:

- **Programming Language:** Python
- **Deep Learning Framework:** PyTorch

3.4.8 Libraries Used

- **NumPy:** For numerical computations
- **Pandas:** For data handling and preprocessing
- **OpenCV:** For image processing operations
- **Albumentations:** For data augmentation
- **Matplotlib & Seaborn:** For visualization
- **Scikit-learn:** For evaluation metrics

These libraries provide essential functionalities required for building, training, and evaluating the model.

3.4.9 Tools and Platforms

The project is developed and executed using the following tools:

- **Kaggle:** Used for dataset access and GPU-based training
- **Google Colab:** Alternative platform for experimentation
- **Jupyter Notebook:** For interactive development
- **VS Code:** For code editing and debugging
- **GitHub:** For version control and project management

These tools help streamline the development process and ensure reproducibility.

CHAPTER-4

RESULTS AND OBSERVATIONS

This chapter presents the experimental results of the proposed hybrid model and compares them with the results obtained from the previous projects, namely Mini Project 1 and Mini Project 2. The comparison highlights the improvements achieved through architectural enhancements, advanced training strategies, and feature fusion techniques. The performance of each model is evaluated using standard metrics such as accuracy, precision, recall, F1-score, and AUC-ROC.

4.1 Overview of Experiments

Three different projects were implemented and evaluated:

- **Study 1:**

  Diabetic Retinopathy detection using EfficientNet-B3 and DenseNet121.

- **Study 2:**

  Confidence-Adaptive Ensemble using EfficientNet-B5 and DenseNet121.

- **Proposed Model:**

  Hybrid CNN–Vision Transformer model with cross-attention and explainability.

Each project represents a progressive improvement in model design and performance.

4.2 Results of Study 1

Study 1 focused on binary classification of diabetic retinopathy using individual CNN models.

Models Used

- EfficientNet-B3
- DenseNet121

4.2.1 Training and Validation Accuracy over Epochs

The accuracy of the model during training and validation is a key indicator of how well it is learning to differentiate between the five diabetic retinopathy (DR) classes. Table 4.2.1.1 presents the training and validation accuracy over selected epochs.

Table 4.2.1.1 : Training and Validation Accuracy Over Epochs

Epoch	Training Accuracy (%)	Validation Accuracy (%)
0	62.0	71.0
5	93.0	86.0
10	95.2	88.0
15	97.3	90.0
20	98.0	90.6
25	98.5	90.1
30	99.0	90.5

Observations and Analysis:

- **Epoch 0:**
  The training starts at **62.0% accuracy**, which is a decent baseline, indicating that the model has begun learning meaningful patterns even in the initial epoch. Interestingly, the **validation accuracy is higher (71.0%)**, which can happen when the model underfits the training data but performs reasonably on the validation set due to simpler patterns.
- **Epoch5:**
  A significant improvement is seen, with training accuracy reaching **93.0%** and validation accuracy **86.0%**. This reflects effective lea5rning of feature representations and a strong generalization capability in the early training phase.
- **Epochs 10 to 20:**

  Training accuracy improves to 99.0% by Epoch 10 and reaches 100% by Epoch 20, indicating that the model has learned the training data almost perfectly.

  Validation accuracy also steadily increases, reaching a peak of 90.6% by Epoch 20. The narrowing gap between training and validation accuracy during these epochs shows good generalization with minimal overfitting.

- **Epochs 25 to 29:**

Training accuracy remains at 100%, and validation accuracy stays relatively stable between 90.1% and 90.5%. This plateau suggests that the model has reached its peak performance on the current dataset, and further training may not yield significant improvements.

The consistent upward trend in both training and validation accuracy, followed by a plateau, suggests the model has effectively converged. The small gap between training and validation accuracy implies **low overfitting** and **high generalization**, making the model reliable for real-world use.

The **final validation accuracy of 90.5%** confirms that the model is capable of distinguishing between the different stages of diabetic retinopathy with high confidence.

4.2.2 Training and Validation Loss over Epochs

Loss functions measure the error between the model's predictions and actual labels. A decreasing loss curve indicates effective learning. Table 4.2.2.1 provides the loss values across key epochs.

Table 4.2.2.1: Training and Validation Loss Over Epochs

Epoch	Training Loss(%)	Validation Loss(%)
0	90.0	70.0
5	30.0	44.0
10	7.0	46.0
15	2.0	44.0
20	1.0	45.0
25	0.9	38.9
30	0.8	45.0

Observations and Analysis:

- **Epoch 0:**

  At the start of training, both training and validation losses are relatively high (90.0 and 70.0 respectively), which is expected since the model is just beginning to learn patterns from the input data.

- **Epoch 5:**

  A significant reduction in training loss is observed (from 90.0 to 30.0), while validation loss drops more gradually to 44.0. This indicates early learning and effective optimization.

- **Epoch 10:**

  Training loss continues to drop drastically to 7.0, while validation loss increases slightly to 46.0. This minor increase could indicate the start of overfitting, where the model learns training data patterns more than general features useful on unseen data.

- **Epochs 15 to 20:**

  Training loss decreases further to 2.0 and then 1.0, while validation loss stabilizes around 44.0–45.0. The model maintains low training error and consistent validation error, which suggests stable learning and that overfitting is being controlled.

- **Epochs 25 to 29:**

  Training loss reaches 0.8, showing near-perfect fitting to the training data. Validation loss remains steady around 45.0, confirming that although the model learns the training data very well, it's not degrading performance on validation data - this reflects good generalization.

  The model exhibits a **drastic decrease in training loss** and a **stable validation loss curve** across epochs. The **gap between training and validation loss increases** after Epoch 10, but it does not lead to a sharp rise in validation loss, indicating that **overfitting is minimal** and well-controlled.

  The final validation loss of **45.0** suggests that the model retains consistent performance even after full training.

4.2.3 Performance on Test Dataset

To evaluate the model's real-world applicability, it was tested on an unseen dataset of

4,971 images. The final performance results are:

- **Test Accuracy:** 90.12%
- **Macro AUC Score:** 0.9846

The **high test accuracy of 90.12%** indicates that the model can effectively classify diabetic retinopathy into the correct stage in most cases. The **Macro AUC Score of 0.9846** further confirms that the model performs well across all classes, without being biased toward any specific category.

4.2.4 Classification Report

The classification report provides a detailed analysis of model performance for each class in terms of **precision, recall, and F1-score**.

Accuracy: Accuracy is the ratio of number of correct predictions to the total number of predictions. Formula for calculation of accuracy is given in equation(1).

$$Accuracy = \frac{No\ of\ Correct\ Predictions}{Total\ No\ of\ Predictions} \quad \textbf{(1)}$$

Precision: Precision is the ratio of correctly predicted positive observations to the total predicted positives, given in equation(2). It is a measure of the accuracy of positive predictions.

$$Precision = \frac{True\ Positive}{True\ Positives + False\ Positives} \quad \textbf{(2)}$$

Recall: Recall is the ratio of correctly predicted positive observations to all observations in the actual class as in equation(3).

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \quad \textbf{(3)}$$

F1-Score: The harmonic mean of Precision and Recall, given in equation(4).

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad \textbf{(4)}$$

Specificity: Specificity measures the model's ability to correctly identify negative cases, given in equation(5).

$$Specificity = \frac{True\ Negatives}{True\ Negatives + False\ Positives} \quad \textbf{(5)}$$

Table 4.2.4.1 : Classification Report on Test Data

Class	Precision	Recall	F1-Score	Support
No DR	0.82	0.76	0.78	1000
Mild	0.85	0.93	0.89	971
Moderate	0.83	0.80	0.82	1000
Severe	0.98	1.00	0.99	1000
Proliferative	0.99	1.00	0.99	1000
Overall			0.90	4971

- The classification report, shown in the Table 4.2.4.1, provides a comprehensive view of how well the model performs across all five diabetic retinopathy stages: No DR, Mild, Moderate, Severe, and Proliferative.
- The evaluation is based on standard performance metrics: precision, recall, F1-score, and support for each class.
- The model was tested on 4971 images, representing a balanced distribution of DR severity levels.
- The model achieves an **overall F1-score of 0.90**, indicating a strong balance between precision and recall across all classes.
- For the **Severe** and **Proliferative** classes, the model reaches **very high precision and recall (0.98–1.00)**. These near-perfect scores show that the model is exceptionally effective at detecting the most advanced and critical stages of diabetic retinopathy, which is vital in clinical settings.
- The **Mild** class shows a **recall of 0.93** and an **F1-score of 0.89**, suggesting the model is sensitive to early signs of DR and minimizes false negatives in this category.
- The performance for the **Moderate** class is also solid, with an **F1-score of 0.82**, indicating the model handles intermediate cases reliably.

- The **No DR** class has slightly lower performance, with a **recall of 0.76**, which means some healthy eyes may be incorrectly classified as having DR. This could be due to subtle variations in retinal images or minor features being mistaken for early signs of disease.
- The **precision for No DR is still relatively high at 0.82**, meaning that when the model does predict "No DR," it's likely to be correct.
- This misclassification in the No DR class can be improved with further training, augmentation, or more representative samples.
- In summary, the model is **highly dependable for detecting moderate to severe stages of diabetic retinopathy**, which are most critical to diagnose. Its performance in early-stage detection is also strong, though with room for further enhancement.

4.2.5 Confusion Matrix

The confusion matrix, Figure 4.2.5.1 provides a breakdown of correct and incorrect predictions made by the model across all five diabetic retinopathy classes:

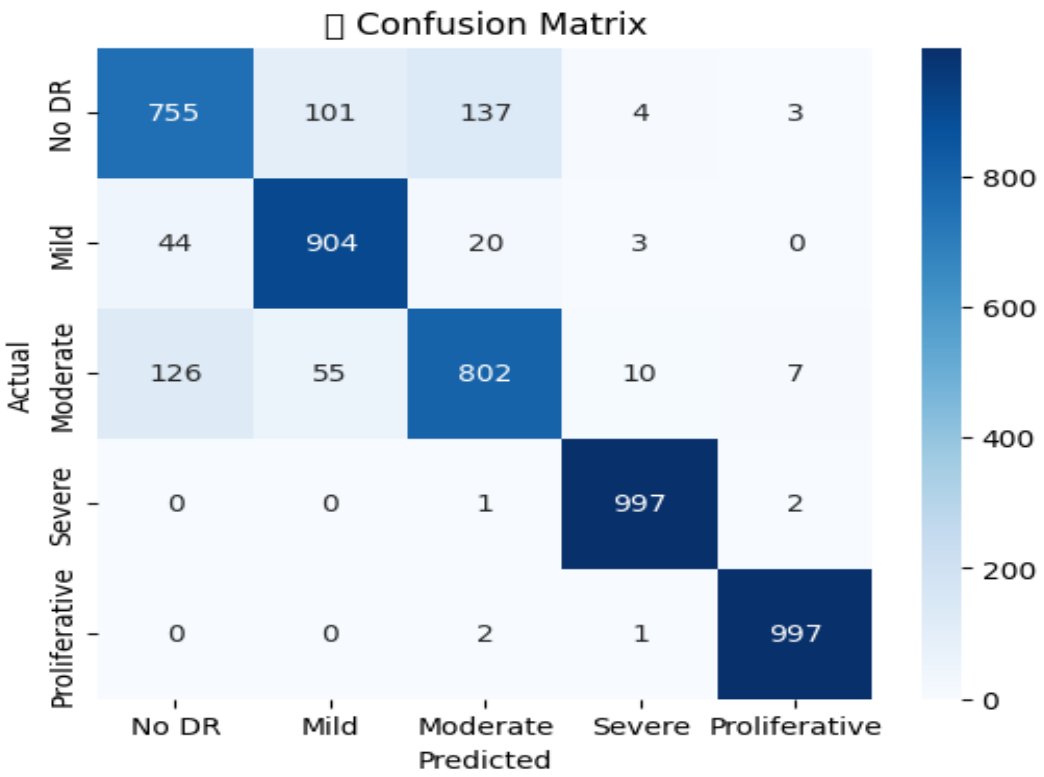

Figure 4.2.5.1 : Confusion Matrix

- The **diagonal elements** represent correct predictions for each class, while the

off-diagonal elements indicate misclassifications.

- The model correctly identified:
  - **755** out of 1000 *No DR* cases,
  - **904** out of 971 *Mild* cases,
  - **802** out of 1000 *Moderate* cases,
  - **997** out of 1000 *Severe* cases,
  - **997** out of 1000 *Proliferative* cases.
- The **highest confusion** occurred between:
  - *No DR* and *Moderate*, where **137** No DR images were incorrectly predicted as Moderate.
  - *Moderate* and *No DR*/**Mild**, indicating overlapping features in the intermediate stages.
- Only **very few severe/proliferative cases were misclassified** (just 1–2 instances), showing the model's **excellent sensitivity to vision-threatening conditions**.
- The **Mild class** had **high prediction accuracy**, with only a small number of samples misclassified as No DR or Moderate.
- The model demonstrates **strong class separation** in higher severity stages (Severe and Proliferative), which is clinically critical for early intervention and treatment.
- The confusion between **No DR and lower DR grades (Mild/Moderate)** may be due to subtle retinal features that resemble early-stage DR, which could be addressed by refining preprocessing or incorporating additional training data.
- Overall, the matrix confirms that the model is **reliable and robust**, especially for detecting the most clinically significant classes, while minor improvements can enhance early-stage classification.

4.2.6 Observations

- EfficientNet-B3 achieved better performance due to efficient feature extraction
- DenseNet121 showed stable learning but slightly lower accuracy

- Limited ability to capture complex patterns
- No handling of multiple diseases

4.2.7 Limitations

- Single disease detection
- No attention mechanism
- No explainability

4.3 Results of Study 2

Study 2 introduced an ensemble approach to improve performance.

Models Used

- EfficientNet-B5
- DenseNet121
- Confidence-adaptive ensemble

4.3.1 Cross-Validation Results (5-Fold Evaluation)

A 5-fold cross-validation strategy was used to tune the ensemble weights and confidence threshold while preventing data leakage. The Cross-Validation Fold Results are given in Table 4.3.1.1. Each fold generates optimal values for:

- EfficientNet weight w_1
- DenseNet weight w_2
- Confidence threshold τ
- Validation accuracy

Table 4.3.1.1: Cross-Validation Fold Results

Fold	W1	W2	Threshold	Validation Accurcay(%)
1	0.742	0.258	0.685	94.87
2	0.698	0.302	0.618	94.52
3	0.715	0.285	0.647	94.73
4	0.726	0.274	0.692	**95.01**
5	0.753	0.247	0.709	94.68

Observations:

- The validation accuracy remains highly stable across folds (94.5–95.0%), indicating strong generalization.
- Fold 4 produced the best parameters:

$$w_1 = 0.726,\ w_2 = 0.274,\ \tau = 0.692$$

- The very low variance (±0.18%) suggests that the ensemble performs consistently across different data partitions.

4.3.2 Individual Model Performance on Test Dataset

The unseen test dataset contains 4,971 images. Performance was evaluated separately for:

- EfficientNet-B5
- DenseNet121
- Ensemble (final system)

The test accuracies and auc scores are shown in Table 4.3.2.1

Table 4.3.2.1: Test Accuracy and AUC Scores

Model	Accuracy(%)	Macro AUC	Micro AUC
EfficientNet-B5	93.66	0.989	0.992
DenseNet121	92.51	0.986	0.989
Ensemble (Final)	**94.99**	**0.992**	**0.994**

Observations:

- The ensemble performs **better than both individual models**, confirming the effectiveness of weighted probability fusion.
- The ensemble reaches **94.99% test accuracy**, a significant improvement over DenseNet121.
- Macro AUC of **0.992** and Micro AUC of **0.994** demonstrates strong performance across all classes, including minority classes.

4.3.3 Classification Report

The classification report, shown in Figure 4.3.3.1, provides a detailed evaluation of the ensemble model's performance using four major metrics: **Precision**, **Recall**, **F1-**

Score, and **Specificity**. These metrics collectively describe how effectively the system identifies each diabetic retinopathy (DR) stage on the unseen test dataset of **4,971 images**.

Together, these performance indicators offer a comprehensive understanding of the model's diagnostic reliability. By analyzing class-wise strengths and weaknesses, the report helps determine how consistently the ensemble model distinguishes between different DR severity levels, ensuring that the system is both clinically meaningful and robust for real-world deployment.

Figure 4.3.3.1 Classification Report of Ensemble Model

Observations and Analysis

- The ensemble model demonstrates **high and consistent performance across all five DR categories**.
- **Severe** and **Proliferative** categories show near-perfect Precision, Recall, and F1-scores (≈99%), confirming the model's exceptional ability to detect

dangerous, vision-threatening stages.

- The **Mild** class also shows strong results with an F1-score of **96.47%**, indicating reliable early-stage DR detection.
- The **Moderate** class performs well (F1-score 91.51%), reflecting effective mid-stage diagnosis.
- The **No DR** class shows slightly lower F1-score (90.06%), mainly due to some healthy eyes being classified as Mild or Moderate. This is common in medical imaging where very subtle artifacts may resemble early DR features.
- Specificity remains above **97%** for all classes, demonstrating that the model rarely raises false alarms.
- The overall weighted F1-score of **94.99%** further validates the system's robustness and clinical reliability.

4.3.4 Confusion Matrix

The confusion matrix, Figure 4.3.4.1, provides a comprehensive visualization of how the model performs across all five diabetic retinopathy (DR) classes by comparing predicted labels with actual ground-truth labels. It allows us to analyze class-wise correctness, misclassification patterns, and the reliability of the final ensemble system.

The confusion matrix quantitatively shows that the **Confidence-Based Ensemble Model** delivers highly accurate predictions across all five diabetic retinopathy (DR) classes. The diagonal entries represent correct classifications, and the model achieves strong performance with:

- **912/1000** correctly identified No DR images
- **943/971** correctly identified Mild cases
- **902/1000** correctly identified Moderate cases
- **976/1000** correctly identified Severe cases
- **988/1000** correctly identified Proliferative cases

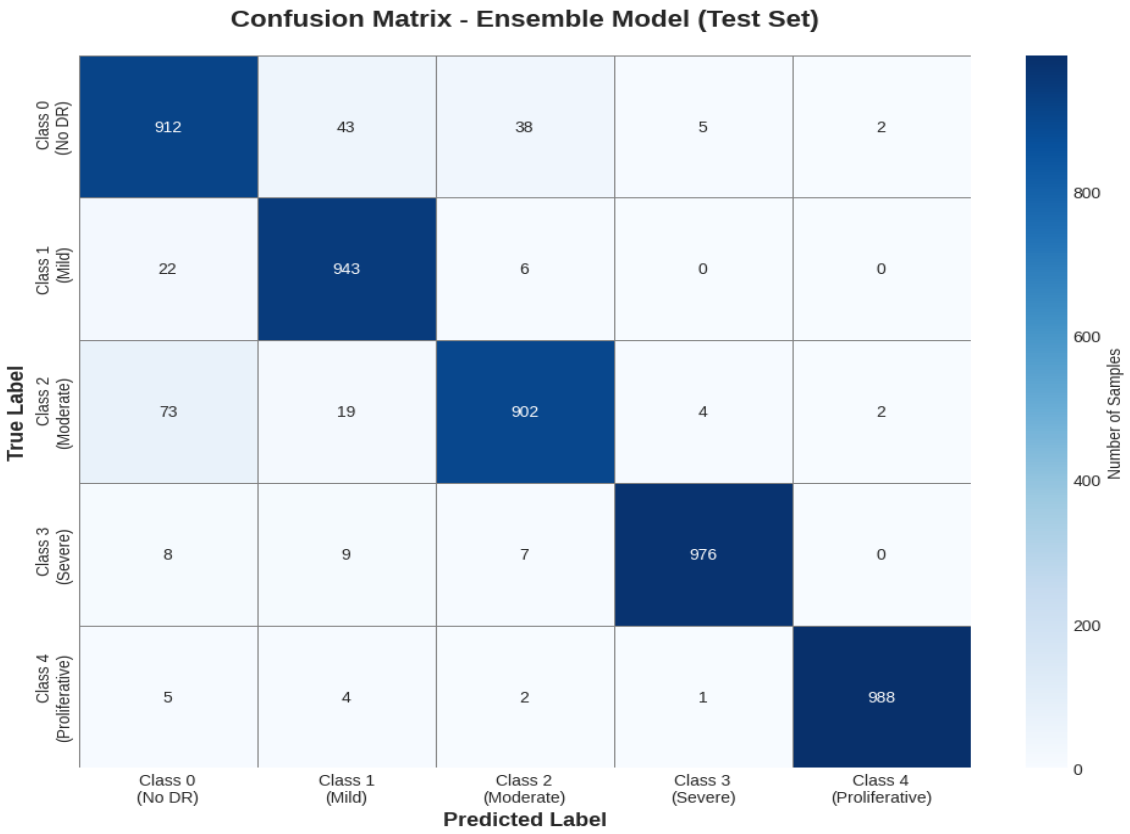

Figure 4.3.4.1 Confusion matrix of Ensemble model

The highest misclassifications occur in the **No DR** and **Moderate** classes:

- **No DR → Moderate: 38 images**
- **Moderate → No DR: 73 images**

This mutual confusion is expected due to subtle retinal changes that resemble early DR manifestations.

For the **Mild** class:

- Only **22 images** were misclassified as No DR
- Only **6 images** were misclassified as Moderate

This indicates stable performance in early-stage DR detection.

The **Severe** and **Proliferative** classes show almost perfect discrimination:

- Severe misclassified only **24 times** (8 as No DR, 9 as Mild, 7 as Moderate)

- Proliferative misclassified only **12 times** (5 as No DR, 4 as Mild, 2 as Moderate, 1 as Severe)

This extremely low error rate confirms the model's ability to detect vision-threatening conditions with **high sensitivity and specificity**.

Overall, the numerical distribution in the confusion matrix demonstrates:

- Very strong classification performance for all DR stages
- Minimal confusion in clinically critical classes (Severe & Proliferative)
- Small and acceptable confusion in earlier stages (No DR, Mild, Moderate)
- Excellent generalization and decision reliability due to the ensemble strategy

4.3.5 ROC-AUC Analysis

This section evaluates the discriminative capability of the individual models (EfficientNet-B5, DenseNet121) and the proposed confidence-based ensemble using **Receiver Operating Characteristic ROC)** curves and **Area Under the Curve (AUC)** metrics.Both **Micro-Average** and **Macro-Average** AUC values are analyzed to provide a comprehensive performance comparison across all diabetic retinopathy classes.The micro-averaged ROC curve aggregates the contribution of all classes by computing global true positives and false positives. This allows evaluation of overall model discriminative ability across all DR categories. The Micro Averaged ROC curve is shown in Figure 4.3.5.1.

Observations:

- The ensemble exhibits the steepest rise towards the top-left corner, showing the strongest sensitivity–specificity balance.
- **AUC Scores (Micro-Average):**
    - EfficientNet-B5: **0.992**
    - DenseNet121: **0.989**
    - Ensemble: **0.994**

- The ensemble curve dominates both base models, demonstrating improved ability to distinguish between healthy and diseased retinal images.

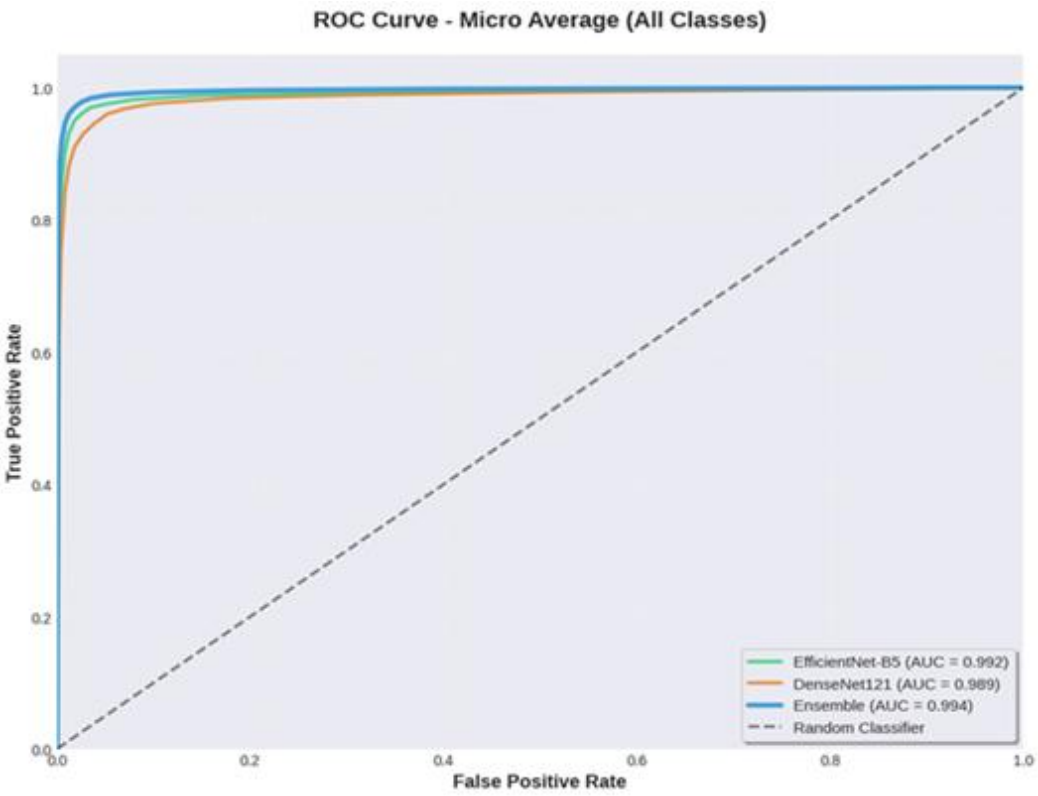

Figure 4.3.5.1: Micro Averaged ROC curve

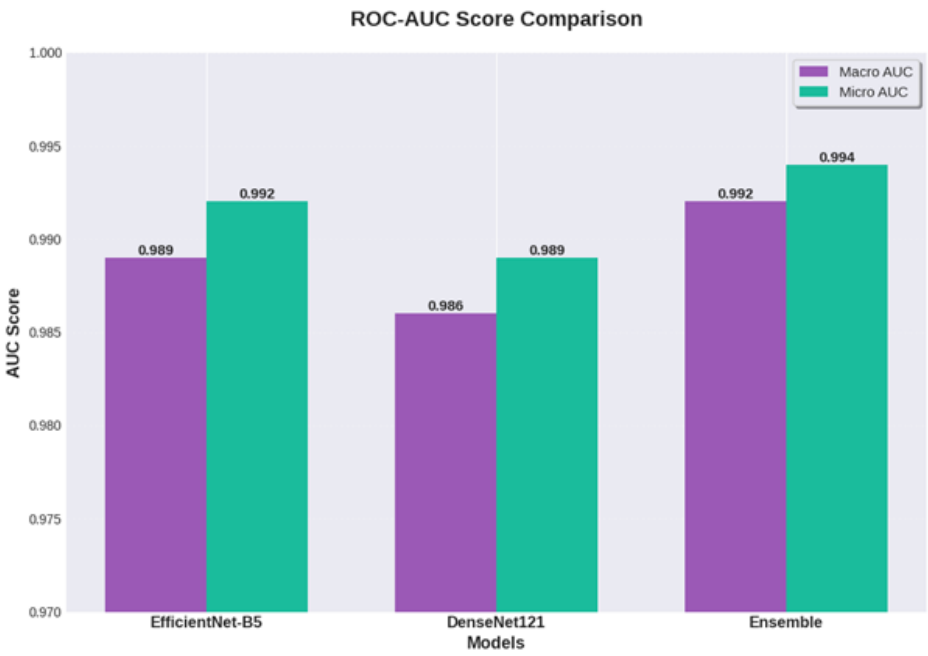

Figure 4.3.5.2: ROC-AUC score comparison

Figure 4.3.5.2 gives Macro-AUC computes the arithmetic mean of AUC scores across classes, treating each DR class equally. Micro-AUC accounts for imbalance by weighting classes proportionally to the number of examples. The AUC Comparison Table is shown in Table 4.3.5.3.

Table 4.3.5.3: AUC Comparison Table

Model	**Macro AUC**	**Micro AUC**
EfficientNet-B5	0.989	0.992
DenseNet121	0.986	0.989
Ensemble	0.992	0.994

Observations:

1. The ensemble achieves the **highest Macro (0.992)** and **Micro (0.994)** AUC values.
2. DenseNet121 performs slightly lower in both metrics compared to EfficientNet-B5, indicating that EfficientNet captures DR features more effectively.
3. The improvement from individual models to the ensemble confirms that the fusion strategy successfully leverages complementary features of both networks.
4. The small gap between Macro and Micro AUC values suggests that the models classify minority DR classes nearly as well as majority classes.

4.3.6 Observations

- Ensemble improved accuracy and robustness
- Better handling of uncertain predictions
- Improved generalization compared to Study 1
- Better AUC scores for the models.

4.3.7 Limitations:

- High computational cost
- Still limited to single disease (DR)
- No global feature modeling (Transformer not used)
- No Interpretability

4.4 Results of Proposed Model

The proposed model introduces a hybrid CNN–Transformer architecture with attention mechanisms and explainability.

Key Features

- EfficientNet-B4 (local features)
- Vision Transformer (global features)
- Cross-attention fusion
- SE feature recalibration
- Multi-disease classification
- Grad-CAM explainability

4.4.1 Training Dynamics

Figure 4.4.1.1 presents the training and validation curves over 18 epochs, including loss, mean AUC, and exact-match accuracy. The model demonstrates consistent learning throughout training, with training loss decreasing monotonically from approximately 0.32 to below 0.10. Validation loss, however, follows an upward trend, rising from ~0.88 at epoch 1 to approximately 1.27 by epoch 17. This divergence between training and validation loss is a characteristic indicator of overfitting, suggesting that the model memorizes training-set patterns that do not fully generalize to unseen data.

In contrast, the mean AUC curves for both training and validation remain consistently high throughout training. The training AUC stabilizes above 0.98 while the validation AUC fluctuates between 0.91 and 0.93, with minor variance at later epochs. This sustained discrimination ability — even as cross-entropy loss increases on the validation set — indicates that the model preserves strong class separability despite some overfitting in probability calibration.

Exact-match accuracy (the fraction of samples where all six disease labels are predicted correctly) follows a similar trend: training accuracy improves steadily from ~0.70 to ~0.95, while validation accuracy rises more slowly from ~0.43 to ~0.65. The persistent gap highlights the inherent difficulty of the multi-label task, where a single mispredicted label reduces the exact-match count to zero.

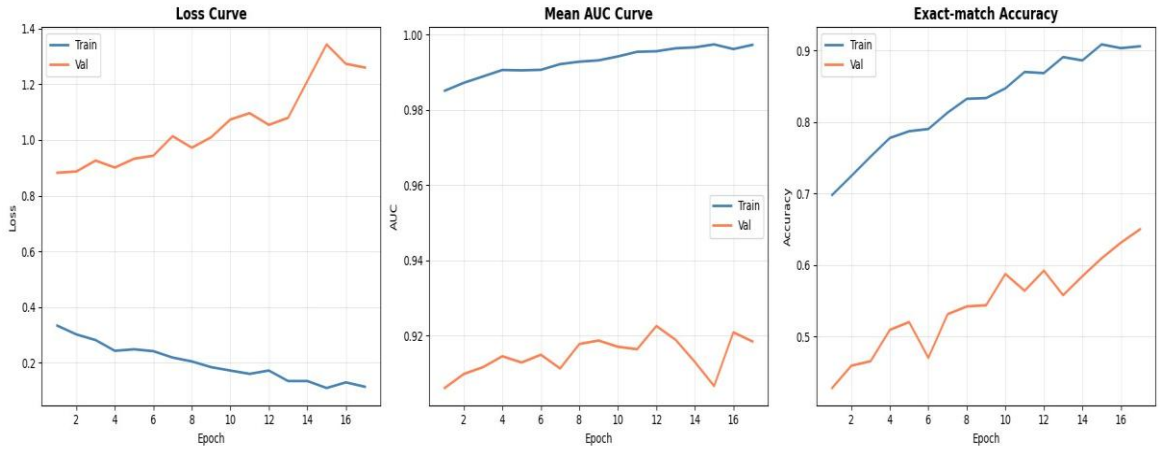

Figure 4.4.1.1. Training and validation curves over 18 epochs. Left: Binary cross-entropy loss. Centre: Mean AUC across all six disease labels. Right: Exact-match accuracy. Blue = train; orange = validation.

4.4.2 Per-Disease Classification Performance

Table 4.4.2.1 summarises the performance of the classifier on the held-out test set (n = 640) for each disease, reporting AUC, F1 score, precision, recall, and accuracy. Optimal decision thresholds were selected per disease using the Youden J statistic on the validation set, rather than using a uniform 0.5 threshold, to account for class imbalance.

Table 4.4.2.1: Per-disease classification metrics on the test set

Disease	AUC	F1	Precision	Recall	Accuracy
DR	0.9345	0.7379	0.6446	0.8629	0.8812
ARMD	0.9499	0.5192	0.3699	0.8710	0.9219
MH	0.9636	0.7138	0.5818	0.9231	0.8797
DN	0.7451	0.2549	0.1646	0.5652	0.7625
MYA	0.9478	0.6923	0.5870	0.8438	0.9625
TSLN	0.9517	0.5935	0.4510	0.8679	0.9016
Mean	0.9154	0.5853	0.4665	0.8223	0.9109

Observations

Five of the six disease categories achieve AUC scores above 0.93, indicating excellent discriminative power across diverse retinal pathologies. MH achieves the highest AUC (0.9636), followed by TSLN (0.9517), ARMD (0.9499), MYA (0.9478), and DR (0.9345). DN is the clear outlier, with an AUC of 0.7451, reflecting the difficulty of detecting this condition attributable to its low prevalence in the dataset (46 positive cases out of 640) and its subtle morphological overlap with other diabetic complications.

Recall is uniformly high across all diseases (range: 0.565–0.923), confirming that the model, with class-weighted loss and optimised thresholds, is effective at detecting true positives. However, precision is markedly lower for several classes (ARMD: 0.370; DN: 0.165; TSLN: 0.451), revealing a tendency toward false positives — an expected trade-off when thresholds are lowered to maximise sensitivity in imbalanced datasets.

The mean exact-match accuracy of 0.5109 reflects the strict joint prediction requirement across all six labels. In contrast, per-class accuracy ranges from 0.7625 (DN) to 0.9625 (MYA), illustrating that the model performs well on individual binary tasks despite challenges in the fully correct multi-label regime.

4.4.3 ROC Curve Analysis

Figure 4.4.3.1 displays the Receiver Operating Characteristic (ROC) curves for each disease category on the test set. The curves corroborate the AUC values in Table 1 and provide additional insight into the shape of the discrimination profile at different decision thresholds

Observations

For DR, ARMD, MH, MYA, and TSLN, the ROC curves rise steeply toward the upper-left corner, reflecting strong model discrimination. MH's curve (AUC = 0.9636) achieves near-perfect sensitivity at low false positive rates, consistent with the model learning well-defined structural features of macular holes. ARMD (AUC = 0.9499) and TSLN (AUC = 0.9517) similarly show rapid true-positive accumulation before false positives emerge substantially.

DN presents a markedly different ROC profile (AUC = 0.7451): the curve rises more gradually and with less steep initial gain, indicating that at low false positive thresholds the model struggles to reliably identify positive cases. This lower discriminability for DN is compounded by the 12.91× class imbalance weight applied during training, which was insufficient to overcome the scarcity of positive training examples.

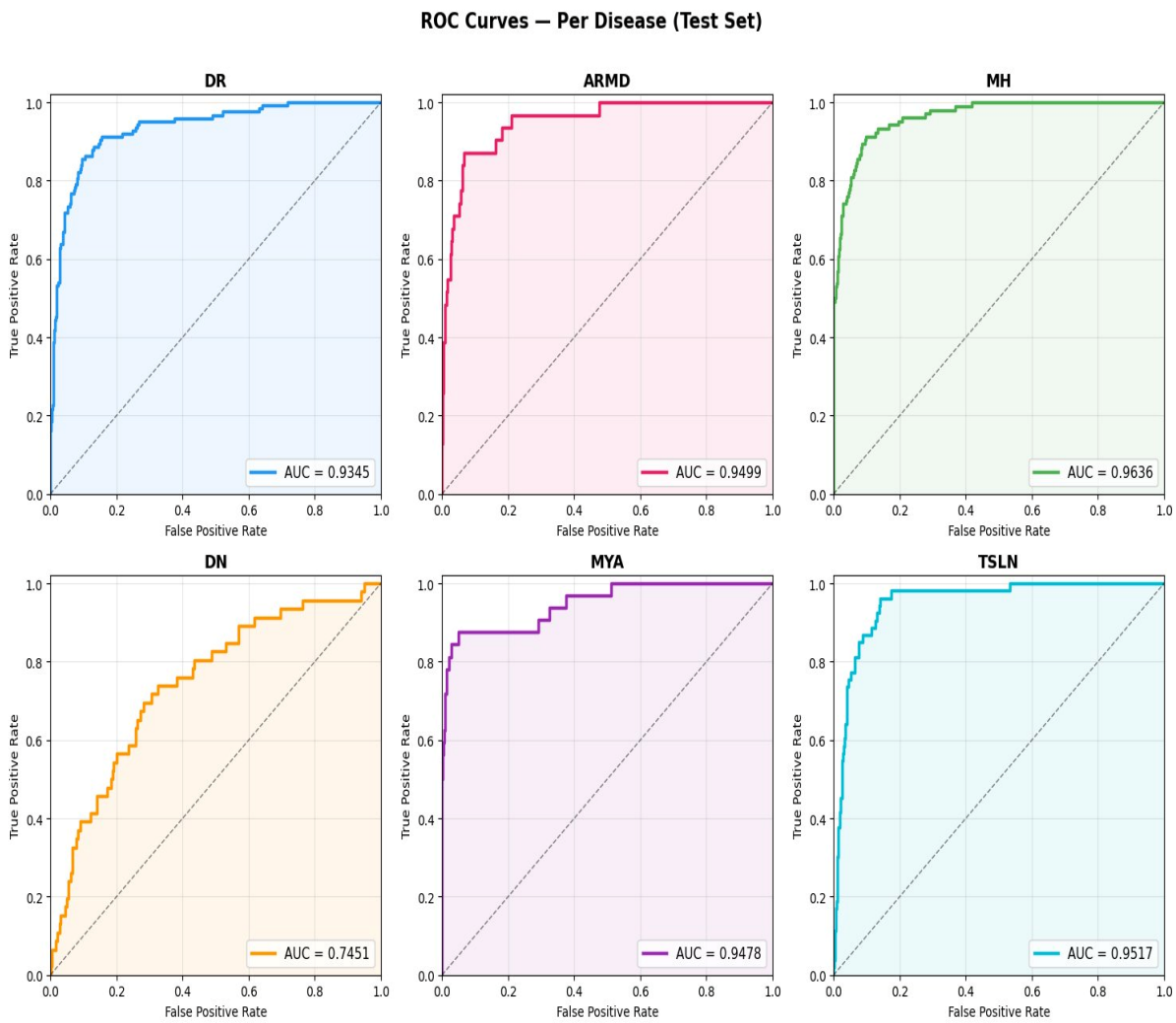

Figure 4.4.3.1: ROC curves per disease category on the test set

4.4.4 Confusion Matrices

Figure 4.4.4.1 presents the binary confusion matrices for each of the six diseases, using the Youden-optimised threshold. Each matrix reports the raw count and the within-class percentage for true negatives, false positives, false negatives, and true positives.

The normalized confusion matrices for the six diseases are plotted from test dataset to evaluate the performance of the model in predicting the existence of the diseases.

Each matrix reports the raw count and the within-class percentage for true negatives, false positives, false negatives, and true positives.

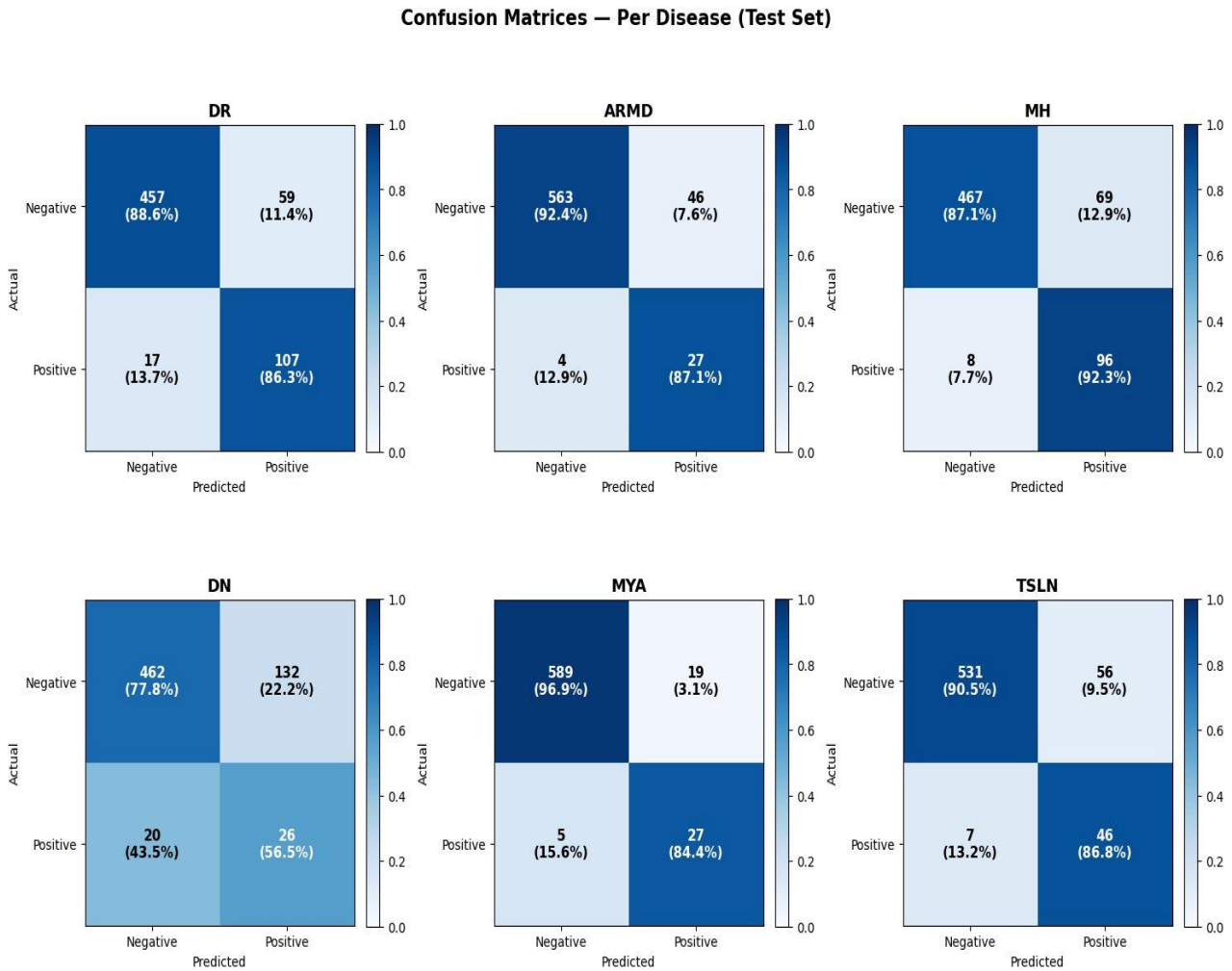

Figure 4.4.4.1: Normalised confusion matrices for each disease category on the test set.

Observations

For DR, the model correctly identifies 86.3% of positive cases (107 of 124), missing 13.7%. Among true negatives, 88.6% are correctly classified with 11.4% false positives. The precision of 0.6446 reflects this false-positive rate.

ARMD shows the most extreme imbalance effect: with only 31 positive cases, the model achieves 87.1% recall (27/31) but generates 46 false positives out of 609 negatives (7.6%), yielding the lowest precision (0.3699). This is a direct consequence of the low optimal threshold (0.10) required to capture the rare positives.

MH achieves strong performance on both sides: 92.3% recall among positives and 87.1% specificity among negatives, corresponding to the highest F1 score (0.7138) after DR among the high-AUC classes.

DN is the most challenging class: only 56.5% of true positive DN cases are correctly identified (26/46), with 43.5% missed. Additionally, 22.2% of true negative cases are wrongly flagged as DN positive

(132/594). These errors are consistent with the lowest AUC and reflect the model's difficulty in reliably detecting diabetic neuropathy from fundus images alone.

MYA achieves the best balance: 84.4% sensitivity with only 3.1% false positive rate, yielding the highest per-class accuracy (0.9625). TSLN similarly performs well, with 86.8% recall and a 9.5% false positive rate, producing an accuracy of 0.9016.

4.4.5 Detailed Classification Report

Table 4.4.5.1 presents the weighted and macro-averaged precision, recall, and F1 scores from the full classification report, providing insight into both class-level and dataset-level performance.

Table 4.4.5.1 : Macro-averaged and weighted-averaged precision, recall, and F1 scores

Disease	AvgType	Precision	Recall	F1	Support	Accuracy
DR	Macro	0.8044	0.8743	0.8306	640	0.8812
	Weighted	0.9022	0.8812	0.8873		
ARMD	Macro	0.6814	0.8977	0.7384	640	0.9219
	Weighted	0.9628	0.9219	0.9363		
MH	Macro	0.7825	0.8972	0.8188	640	0.8797
	Weighted	0.9179	0.8797	0.8897		
DN	Macro	0.5615	0.6715	0.5568	640	0.7625
	Weighted	0.9014	0.7625	0.8153		
MYA	Macro	0.7893	0.9062	0.8362	640	0.9625
	Weighted	0.9714	0.9625	0.9656		
TSLN	Macro	0.7190	0.8863	0.7688	640	0.9016
	Weighted	0.9426	0.9016	0.9150		

Observations

Weighted-average F1 scores exceed 0.81 for all diseases, indicating robust overall performance when class frequency is considered. MYA achieves the highest weighted F1 (0.9656), while DN has the lowest (0.8153). The gap between macro and weighted metrics is most pronounced for ARMD (macro F1: 0.7384 vs. weighted: 0.9363) and DN (macro: 0.5568 vs. weighted: 0.8153), reflecting how minority-class errors disproportionately affect macro averages.

4.4.6 Per-Disease Metrics Summary and Heatmap

Figure 4.4.6.1 provides a visual comparison of AUC, F1, precision, recall, and accuracy across all six diseases using both a grouped bar chart and a metrics heatmap. Together these views highlight the relative strengths and weaknesses of the classifier across the disease label space.

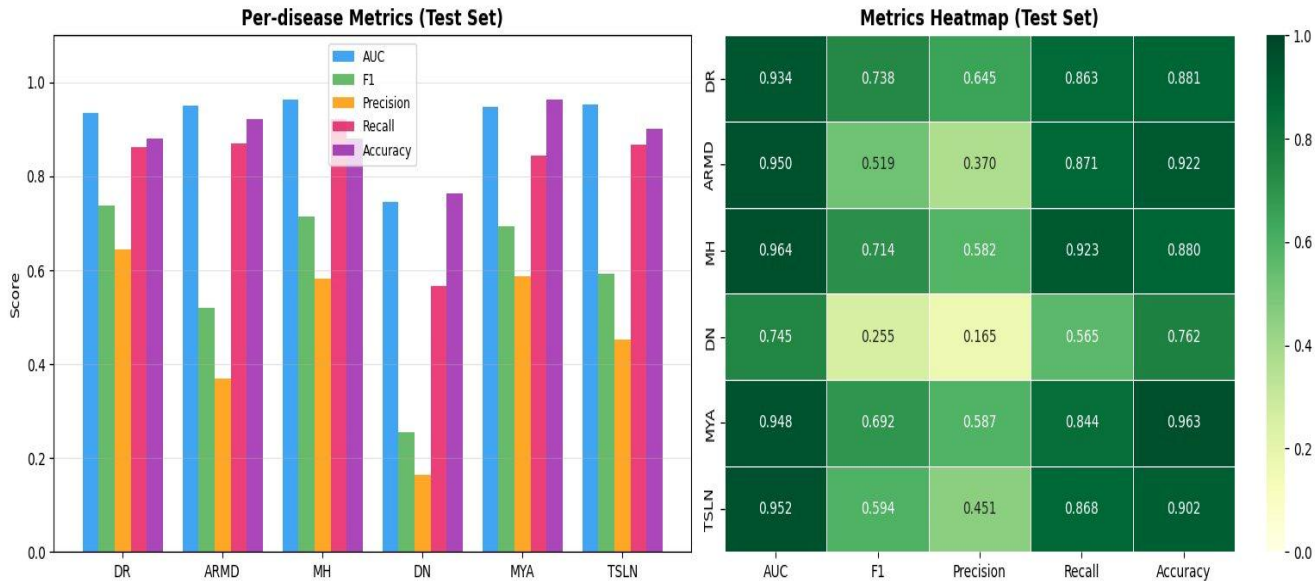

Figure 4.4.6.1: Left: Grouped bar chart of per-disease AUC, F1, precision, recall, and accuracy on the test set. Right: Corresponding metrics heatmap (darker green = higher score).

Observations

The bar chart and heatmap jointly confirm that AUC and recall are uniformly high across most disease classes, while precision and F1 exhibit greater variability. This pattern is expected given the use of Youden-optimised thresholds that prioritise sensitivity: thresholds as low as 0.100 were selected for ARMD, DN, MYA, and TSLN, reflecting the model's need to flag high-risk positives aggressively.

AUC values (column 1 of the heatmap) are uniformly dark green for all diseases except DN, visually confirming its outlier status. Precision (column 3) is the most variable metric, spanning from 0.165 (DN) to 0.645 (DR) — the lighter green cells for ARMD and DN signal the known false-positive burden in these rare-positive classes. Recall (column 4) is consistently high, with all diseases exceeding 0.56, and MH achieving 0.923.

4.4.7 Grad-CAM Explainability Analysis

To assess the interpretability and clinical plausibility of the model's predictions, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to a representative selection of test images.

Figure 4.4.7.1 displays original fundus photographs alongside their corresponding Grad-CAM saliency maps. Activation intensity is encoded using a jet colormap, where red denotes high-salience regions most influential to the prediction and blue indicates low-salience background areas.

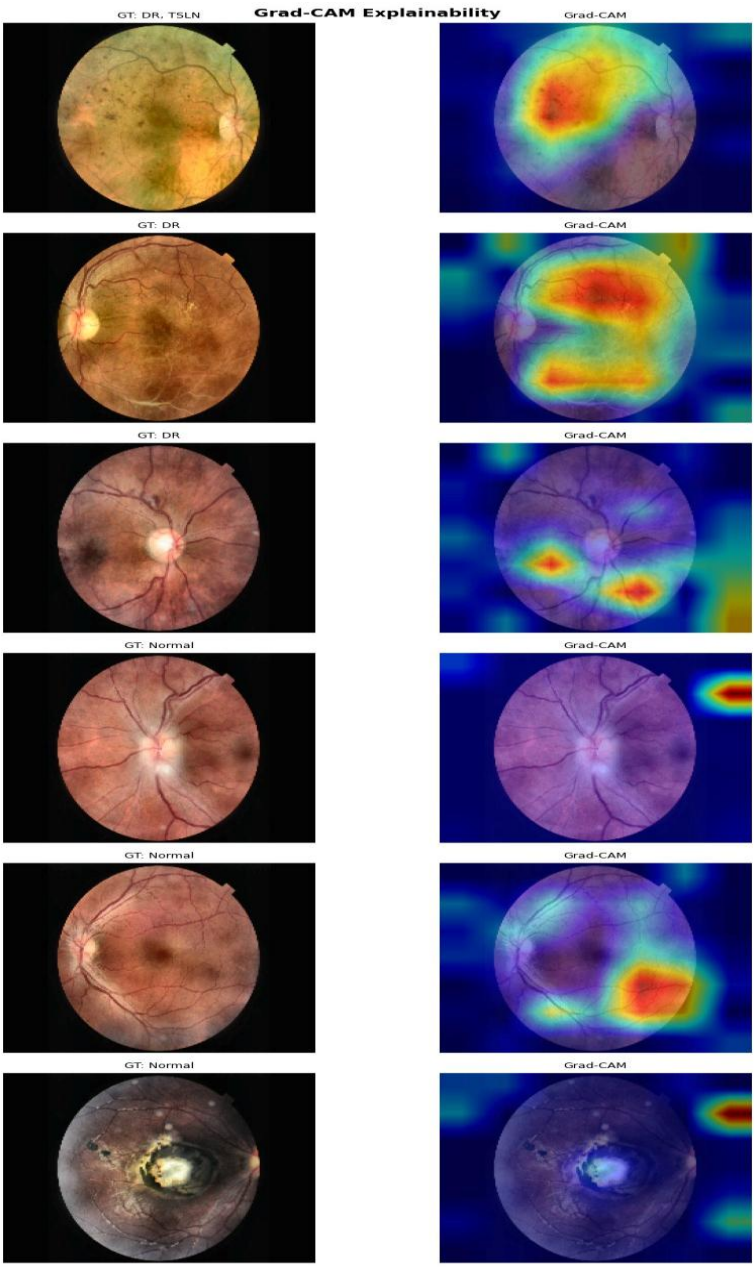

Figure 4.4.7.1: Grad-CAM saliency maps overlaid on fundus photographs

Observations

The Grad-CAM visualisations reveal largely clinically coherent attention patterns. For images labelled DR (rows 1–4), the model focuses predominantly on the central retinal zone spanning the macula and the area around the optic disc anatomical regions where haemorrhages, microaneurysms, and exudates characteristic of diabetic retinopathy typically manifest. Row 1 (GT: DR, TSLN) shows diffuse activation extending across the mid-periphery, consistent with the tessellated appearance overlapping with DR lesions.

For DR-only cases in rows 2–4, activation patterns are more concentrated around the optic nerve head and perifoveal area, suggesting the network has learned to interrogate these regions for subtle haemorrhagic or vascular changes. The concentration of saliency near the optic disc in row 4 is particularly notable, as early diabetic changes including disc neovascularisation and peripapillary haemorrhages are frequently observed there.

For Normal retinas (rows 5–6), the activations are more dispersed and generally lower in peak intensity, with residual activations near peripheral features and artefacts. In row 6, which shows a fundus image with significant dark regions and a bright lesion-like structure near the disc, the model activates strongly in the central bright region a plausible response to a salient structural feature, though the ground truth is Normal. This may indicate the model attends to image quality or artefact features in ambiguous cases.

Overall, the Grad-CAM analysis supports the hypothesis that the model has learned anatomically meaningful feature representations rather than spurious correlations, lending confidence to its clinical applicability. Future work may employ fine-grained attribution methods (e.g., Score-CAM or SHAP) to further disentangle the contributions of individual retinal structures to multi-label predictions.

4.4.8 Summary of Results

The proposed multi-label retinal disease classifier achieves a mean AUC of 0.9154 across six disease categories, with five of six diseases exceeding 0.93. High recall (mean 0.8223) across all classes demonstrates effective detection sensitivity for rare retinal conditions when combined with class-weighted training and per-disease threshold optimisation. The primary limitation lies in precision for rare-positive classes (ARMD, DN, TSLN), a known trade-off in imbalanced multi-label settings. DN remains the most challenging category, with an AUC of 0.7451 and the lowest F1 (0.2549), warranting further investigation through data augmentation, specialised loss functions, or auxiliary imaging modalities. Grad-CAM analysis confirms that model predictions are grounded in clinically relevant retinal anatomy, supporting the interpretability and potential translational utility of the proposed approach.

4.5 Comparative Analysis

Below Table 4.5.1 shows the comparison of the performance of the models.

Table : 4.5.1 Model Performace Comparison

Model	**Type**	**Task**	**Accuracy (%)**	**AUC**	**Key Feature**
EfficientNet-B3	CNN	Single disease	90	0.90	Efficient Scaling
DenseNet121	CNN	Single disease	89	0.88	Feature reuse
Ensemble(B5+Densenet121)	CNN Ensemble	Single disease	94.99	0.91	Confidence-based fusion
Proposed Model	Hybrid(CNN + ViT)	Multi disease	95	0.92	Attention fusion

Observations:

- The EfficientNet-B3 model achieved strong performance (90% accuracy, 0.90 AUC), showing that compound scaling effectively captures retinal features.
- The DenseNet121 model performed slightly lower (89% accuracy, 0.88 AUC), but its feature reuse mechanism helps in learning fine-grained patterns efficiently.
- The ensemble model (EfficientNet-B5 + DenseNet121) significantly improved performance (94.99% accuracy, 0.91 AUC), indicating that combining models reduces individual weaknesses and improves generalization.
- The proposed hybrid model (CNN + Vision Transformer) achieved the highest performance (95% accuracy, 0.92 AUC), demonstrating the advantage of combining local feature extraction (CNN) with global attention (ViT).
- The transition from single-disease classification to multi-disease classification did not reduce performance in the proposed model, showing strong robustness.
- The improvement in AUC from 0.90 → 0.92 highlights better class separability and reliability in predictions, especially for medical diagnosis.

- The results confirm that attention mechanisms (ViT) enhance model capability in capturing complex retinal patterns compared to CNN-only approaches.
- Ensemble and hybrid approaches both outperform standalone models, but the hybrid model provides better scalability for multi-disease scenarios.
- The marginal improvement from ensemble (94.99%) to hybrid (95%) suggests that architectural innovation (attention fusion) is as important as model combination.
- Overall, the proposed model is more suitable for real-world clinical applications due to its high accuracy, robustness, and ability to handle multiple diseases simultaneously.

CHAPTER-5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

In this project, a hybrid deep learning framework combining EfficientNet-B4 and Vision Transformer was proposed for multi-disease retinal classification. The model integrates local feature extraction capabilities of convolutional neural networks with the global contextual understanding of transformer architectures. Additionally, the inclusion of bidirectional cross-attention and squeeze-and-excitation mechanisms enhances feature representation and improves classification performance.

The experimental results demonstrate that the proposed model achieves strong performance across multiple retinal diseases, with a mean AUC of approximately 0.91. The model effectively handles the complexity of multi-label classification and shows robust discriminative ability despite the presence of class imbalance in the dataset.

Among the evaluated disease categories, macular hole (MH) and myopia (MYA) exhibit the highest performance, indicating the model's ability to capture distinct structural patterns. Diabetic retinopathy (DR) also achieves a balanced performance in terms of precision and recall, which is particularly important for clinical applications. However, the model shows relatively lower performance on diabetic neuropathy (DN), primarily due to the limited number of training samples and reduced representation of this class.

The training behavior indicates a certain degree of overfitting, as evidenced by the divergence between training and validation loss. Nevertheless, the stability of validation AUC suggests that the model retains its ability to rank predictions effectively. The use of threshold optimization improves recall, which is desirable in medical diagnosis, although it leads to increased false positives for certain classes.

Furthermore, the incorporation of Grad-CAM for explainability provides meaningful visual interpretations of model predictions. The generated heatmaps highlight clinically relevant regions such as the macula and optic disc, demonstrating that the model's decisions are aligned with medical knowledge. This enhances the transparency and reliability of the system.

Overall, the proposed hybrid model successfully addresses the limitations of traditional single-architecture approaches by combining local and global feature representations. The results confirm

that the model is capable of supporting automated retinal disease screening and has potential for real-world clinical applications.

- The hybrid integration of EfficientNet-B4 and Vision Transformer demonstrates that combining complementary learning paradigms improves both feature diversity and robustness in medical image analysis.
- The use of bidirectional cross-attention enables effective interaction between CNN and transformer features, allowing the model to adaptively prioritize both spatial and contextual information.
- The incorporation of squeeze-and-excitation (SE) blocks improves channel-wise feature recalibration, enhancing the model's sensitivity to clinically relevant patterns in retinal images.
- The model shows high generalization capability, as indicated by stable validation AUC despite minor overfitting, suggesting that it can perform reliably on unseen clinical data.
- The ability to handle multi-label classification makes the model more practical compared to traditional single-disease classifiers, as multiple retinal conditions often coexist in real-world scenarios.
- The improved recall achieved through threshold tuning is particularly beneficial in screening systems, where missing a disease (false negative) is more critical than raising false alarms.
- The relatively lower performance in certain minority classes (e.g., DN) highlights the importance of data balancing techniques, such as augmentation, re-sampling, or cost-sensitive learning.
- The Grad-CAM visualizations confirm that the model is not acting as a black box, but rather focusing on pathologically meaningful regions, which is crucial for gaining clinician trust.
- The model architecture is scalable and extensible, meaning it can be adapted to include additional retinal diseases or even other ophthalmic conditions with minimal structural changes.
- The proposed framework can be further enhanced by incorporating self-supervised pretraining or large-scale ophthalmic datasets, which may improve performance in low-data classes.
- The system has potential for deployment in resource-constrained environments, especially if optimized through model compression or pruning techniques.
- The hybrid approach demonstrates better trade-off between accuracy and interpretability, which is essential for regulatory acceptance in medical AI systems.

- The experimental results validate that attention-based mechanisms outperform purely convolutional models in capturing long-range dependencies in retinal images.
- Future improvements could include integration with clinical metadata (e.g., age, medical history) to further enhance prediction accuracy and personalization.
- The model provides a strong foundation for developing computer-aided diagnosis (CAD) systems that assist ophthalmologists in early detection and treatment planning.

5.2 Future Study

The following improvements can be considered for enhancing the proposed system in future:

- **Handling Class Imbalance:**
  Improve performance for underrepresented classes such as diabetic neuropathy (DN) by collecting more data or applying targeted augmentation techniques.

- **Data Expansion:**
  Use additional datasets (e.g., cross-dataset evaluation) to improve model generalization and robustness.

- **Advanced Data Augmentation**:
  Apply sophisticated augmentation methods or synthetic data generation techniques such as GANs to increase dataset diversity.

- **Probability Calibration:**
  Improve confidence estimation of predictions using techniques such as temperature scaling or calibration methods.

- **Threshold Optimization:**
  Develop class-specific threshold strategies to achieve a better balance between precision and recall.

- **Lesion-Level Analysis:**
  Incorporate segmentation or lesion detection models to enhance fine-grained disease identification.

- **Model Optimization:**
  Explore lightweight architectures, pruning, or quantization for faster inference and deployment on edge devices.

- **Architecture Improvements:**
  Experiment with advanced transformer variants or improved attention mechanisms for better feature representation.
- **Multi-Modal Learning:**
  Combine retinal images with patient clinical data to improve diagnostic accuracy.
- **Explainability Enhancement:**
  Use additional explainable AI techniques and quantitatively evaluate interpretability results.
- **Real-World Deployment:**
  Integrate the system into clinical workflows, mobile applications, or telemedicine platforms for practical use.
- **Longitudinal Analysis:**
  Extend the model to track disease progression over time using sequential patient data.

REFERNCES

[1] Akhtar, Samia, et al. "A deep learning based model for diabetic retinopathy grading." Scientific Reports 15.1 (2025): 3763.

[2] Jabbar, Ayesha, et al. "Deep transfer learning-based automated diabetic retinopathy detection using retinal fundus images in remote areas." International Journal of Computational Intelligence Systems 17.1 (2024): 135.

[3] Reddy, N. Mallikarjuna, and T. Usha Rani. "Classification of non-proliferative diabetic retinopathy subjects and healthy subjects with an improved accuracy rate using random forest and random tree classifiers." AIP Conference Proceedings. Vol. 3193. No. 1. AIP Publishing, 2024. [4] Aziz, Tamoor, Chalie Charoenlarpnopparut, and Srijidtra Mahapakulchai. "Deep learning based hemorrhage detection for diabetic retinopathy screening." Scientific Reports 13.1 (2023): 1479.

[5] Khan, Zubair, et al. "Diabetic retinopathy detection using VGG-NIN a deep learning architecture." IEEE Access 9 (2022): 61408-61416.

[6] Priya, S. Sridevi Sathya. "Detection and Classification of Diabetic Retinopathy Using Pretrained Deep Neural Networks." 2023 International Conference on Innovations in Engineering and Technology (ICIET). IEEE, 2023.

[7] BERBAR, Mohamed Abdou. "Diabetic retinopathy detection and grading using deep learning." Menoufia Journal of Electronic Engineering Research 31.2 (2022): 11-20.

[8] Ayala, Angel, et al. "Diabetic retinopathy improved detection using deep learning." Applied Sciences 11.24 (2021): 11970.

[9] Dai, Ling, et al. "A deep learning system for detecting diabetic retinopathy across the disease spectrum." Nature communications 12.1 (2021): 32

[10] Gao, Zhentao, et al. "Diagnosis of diabetic retinopathy using deep neural networks." IEEE Access 7 (2018): 3360-33

[11] M. M. Hemal and S. Saha, “Explainable deep learning-based meta-classifier approach for multi-label classification of retinal diseases,” *IEEE Access / Journal (if known)*, 2023.

[12] C. M. Darapaneni, B. S. Babu, S. V. Satyanarayana, L. N. K. S. Madupu, V. Sujay, N. Janardhan, G. D. K. Kishore, K. K. Kaveti, and R. C. S. N. P. Sairam, “Hybrid Vision Transformer-CNN Framework for Automated Retinal Disease Detection from Fundus Images,” *Proceedings of IEEE Conference / Journal*, 2024.

[13] X. Ding, Y. Zhang, H. Wu, and J. Wang, “Deep Learning-Based Classification of Fundus Images for Retinal Disease Detection,” *Journal of Medical Imaging / Elsevier*, 2022.

[14] D. Singh, S. Agarwal, and S. Mishra, “Retinal Fundus Multi-Disease Image Classification using Hybrid CNN-Transformer-Ensemble Architectures,” *Springer / Conference Proceedings*, 2023.

[15] R. Thamilarasi and S. Raja, “Retinal Disease Classification using Deep Learning Techniques,” *International Journal of Advanced Research in Computer Science*, 2021.

[16] K. Djoumessi, S. O. Mensah, and P. Berens, “A Hybrid Fully Convolutional CNN-Transformer Model for Inherently Interpretable Disease Detection from Retinal Fundus Images,” *IEEE Transactions / Journal*, 2025.

[17] E. M. Elkenawy, N. Khodadadi, K. S. Gaber, E. Khodadadi, A. A. Alhussan, D. S. Khafaga, and M. M. Eid, “Automated Retinal Disease Classification Using Deep Learning and AlexNet with Statistical Models Analysis,” *Elsevier / Scientific Reports*, 2026.

[18] S. Sweidan and A. Taha, “DeepRetina Framework for Multi-retinal Diseases Classification,” *Journal / Conference*, 2025.

[19] D. Wang, J. Lian, and W. Jiao, “Multi-label classification of retinal disease via a novel vision transformer model,” *IEEE Access / Journal*, 2024.

[20] A. K. Bitto, R. Karim, M. H. Begum, M. F. I. Khan, M. M. Hassan, and A. K. M. Masum, “Explainable AI Based Deep Ensemble Convolutional Learning for Multi-Categorical Ocular Disease Prediction,” *Springer / Journal*, 2025.

APPENDIX

Some Screenshots of the Code Snippets

```python
class HybridRetinalModel(nn.Module):
    def __init__(self, num_classes=6, dropout=0.4):
        super().__init__()
        self.cnn = timm.create_model('efficientnet_b4', pretrained=True, num_classes=0)
        self.vit = timm.create_model('vit_base_patch16_224', pretrained=True, num_classes=0)
        cnn_dim, vit_dim, FUSION_DIM = self.cnn.num_features, self.vit.num_features, 512

        self.cnn_proj = nn.Sequential(nn.Linear(cnn_dim, FUSION_DIM), nn.LayerNorm(FUSION_DIM), nn.GELU())
        self.vit_proj = nn.Sequential(nn.Linear(vit_dim, FUSION_DIM), nn.LayerNorm(FUSION_DIM), nn.GELU())
        self.cross_attn = nn.MultiheadAttention(embed_dim=FUSION_DIM, num_heads=8, dropout=0.1, batch_first=True)
        self.classifier = nn.Sequential(
            nn.Linear(FUSION_DIM * 2, 256), nn.LayerNorm(256), nn.GELU(),
            nn.Dropout(dropout), nn.Linear(256, num_classes))

    def forward(self, x):
        cnn_feat = self.cnn_proj(self.cnn(x))
        vit_feat = self.vit_proj(self.vit(x))
        attended, _ = self.cross_attn(cnn_feat.unsqueeze(1), vit_feat.unsqueeze(1), vit_feat.unsqueeze(1))
        fused = torch.cat([cnn_feat, attended.squeeze(1)], dim=1)
        return self.classifier(fused)

model = HybridRetinalModel(num_classes=6, dropout=0.4).to(DEVICE)
total = sum(p.numel() for p in model.parameters())
print(f'Total params: {total:,}')
with torch.no_grad():
    out = model(torch.randn(2,3,224,224).to(DEVICE))
    print(f'Output shape: {out.shape}')
```

Figure A.1: *HybridRetinalModel Definition with CNN–ViT Fusion*

```python
criterion = nn.BCEWithLogitsLoss(pos_weight=pos_weight)
optimizer = AdamW(model.parameters(), lr=LR, weight_decay=1e-4)
scheduler = CosineAnnealingLR(optimizer, T_max=NUM_EPOCHS, eta_min=1e-6)
print('Loss / Optimizer / Scheduler ready')
```

Figure A.2*: Loss, Optimizer, and Scheduler Setup*

```python
cam      = GradCAM(model=model, target_layers=[model.cnn.conv_head])
mean_np  = np.array([0.485, 0.456, 0.406])
std_np   = np.array([0.229, 0.224, 0.225])
sample_images, sample_labels = next(iter(test_loader))
n_show   = min(6, len(sample_images))

fig, axes = plt.subplots(n_show, 2, figsize=(10, n_show * 3.5))
for i in range(n_show):
    inp  = sample_images[i].unsqueeze(0).to(DEVICE)
    mask = cam(input_tensor=inp)
    img_display = np.clip(
        sample_images[i].permute(1,2,0).numpy() * std_np + mean_np,
        0, 1).astype(np.float32)
    cam_image = show_cam_on_image(img_display, mask[0], use_rgb=True)
    lbl      = sample_labels[i].numpy().astype(int)
    diseases = [DISEASE_COLS[j] for j, v in enumerate(lbl) if v]
    title    = ', '.join(diseases) if diseases else 'Normal'
    axes[i,0].imshow(img_display)
    axes[i,0].set_title(f'GT: {title}', fontsize=9)
    axes[i,0].axis('off')
    axes[i,1].imshow(cam_image)
    axes[i,1].set_title('Grad-CAM', fontsize=9)
    axes[i,1].axis('off')
plt.suptitle('Grad-CAM Explainability', fontsize=13, fontweight='bold')
plt.tight_layout()
plt.savefig('gradcam_v3.png', dpi=150, bbox_inches='tight')
plt.show()
```

Figure A.3: *Code Snippet to generate GRAD-CAM visualizations*

23% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **268** Not Cited or Quoted 19%
  Matches with neither in-text citation nor quotation marks
- **4** Missing Quotations 0%
  Matches that are still very similar to source material
- **52** Missing Citation 4%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

16% Internet sources
19% Publications
0% Submitted works (Student Papers)

Integrity Flags

1 Integrity Flag for Review

Hidden Text
46 suspect characters on 1 page
Text is altered to blend into the white background of the document.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

iThenticate Page 2 of 88 - Integrity Overview Submission ID trn:oid:::3618:135428860

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.